



QUANTUM

20 October 2024

# CARRIER SECURITY

## R82

Administration Guide



CHECK POINT™

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Revision History

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# GSM Overview

This section gives a quick overview of GPRS, UMTS, and LTE.

## A Global System for Mobile Communications

The most widely deployed wireless networks worldwide are those based on Global System for Mobile Communications, or GSM, technology. Formerly known as "Groupe Special Mobile," GSM is a world-wide standard for digital wireless mobile phones. The standard was originated by the European Conference of Postal and Telecommunications Administrations (CEPT) and further developed by the European Telecommunications Standards Institute (ETSI) as a standard for European mobile phones, with the intention of developing an open, non-proprietary standard for adoption world-wide. It has been remarkably successful, with more than one billion people using GSM phones as of early 2004.

The ubiquity of the GSM standard makes intra-nation roaming very common, with international roaming frequently enabled by "roaming agreements" between operators. GSM differs from its predecessors most significantly in that both signaling and speech channels are digital. It has also been designed for a moderate level of security. GSM employs time division multiple access between stations on a frequency duplex pair of radio channels, with slow frequency hopping between channels.

## General Packet Radio Services

General Packet Radio Services, or GPRS, is a GSM extension which allows packet switched data transmission. GPRS has been called 2.5G, as it is viewed as a stepping stone toward pure 3G systems like UMTS/W-CDMA or similar. GPRS is backward compatible with GSM, a fact that eases the migration path for GSM operators, who can gradually upgrade their infrastructure as the GPRS market expands.

From the user's point of view, GPRS is a wireless extension of data networks. It can access multiple types of data networks, such as IP based networks like the public Internet, private intranet, both IPv4 and IPv6 protocols, and X.25 based networks. GPRS upgrades GSM data services providing:

- Point-to-point (PTP) service: internetworking with the Internet (IP protocols) and X.25 networks.
- Point-to-multipoint (PTM) service: point-to-multipoint multicast and point-to-multipoint group calls.
- Short Message Service (SMS): bearer for SMS.

Thus mobile subscribers can receive an array of services, including web browsing, e-mail communications, intranet access and location-based services.

GPRS is basically an addition to GSM that enables packet based communications. Data transmitted by packet switching is faster and more efficient than circuit switching, the method used in 2G networks. Whereas in GSM timeslots are normally allocated to create a circuit-switched connection, in GPRS timeslots are allocated to a packet-connection on an as-needed basis. This means that if no data are sent by a sender, the frequencies involved remain free to be used by others. Users of GPRS networks can stay continuously logged in to email and Internet services, while paying for these services only when sending and receiving data.

Development of GPRS is directed by the 3rd Generation Partnership Project (3GPP), a collaboration agreement established in 1998. 3GPP's original goal was to produce technical specifications for third generation mobile systems, and now is involved in maintaining and developing GSM standards, including GPRS.

## Universal Mobile Telecommunications System

Universal Mobile Telecommunications System, or UMTS, is one of the third generation (3G) mobile phone. UMTS further extends the capabilities of GPRS networks, offering much higher air interface bandwidth. UMTS networks provide an average bandwidth of up to 384Kbit/sec, which is more than 26 times the bandwidth obtainable on a single GSM error-corrected circuit switched data channel. This increased bandwidth allows for the development and support of a whole new set of services, mostly multimedia-based, such as video streaming, video conferencing, online games, advanced location services, and more.

## IP Multimedia Subsystem

A description of the evolving UMTS network would not be complete without mentioning IP Multimedia Subsystem, or IMS. The IP Multimedia Subsystem (IMS) is a common architecture that allows cellular operators to provide multimedia services. Promoted by 3GPP, IMS uses SIP as its basic signaling protocol.

IMS uses SIP to register and authenticate the mobile user when joining a multimedia session, as well as to initiate the session by locating the destination of the session (either a multimedia server, or other mobile user, or other non-mobile user).

By selecting a standard protocol for multimedia services, the aim is to eliminate interoperability issues in the creation of multimedia sessions between mobile users, and between mobile users and users on the Internet.

Check Point's portfolio of cellular security solutions includes solutions for IMS security as well.

# Long Term Evolution (LTE)

In response to the high demand for connectivity for new hand-held devices and mobile applications, the 3rd Generation Partnership Project, a collaboration between telecommunication associations and the European Telecommunications Standards Institute, has introduced the Long Term Evolution specification. LTE, also known as 4G, is a cellular internet protocol designed to increase the speed and download capacity of mobile (wireless) networks.

The LTE term also refers to SAE, or System Architecture Evolution. SAE is the core network architecture of the 3rd Generation Partnership Project's wireless communication standard.

## Basic Components of GPRS/UMTS Networks

### On the Network

PLMN (Public Land Mobile Network) - a mobile wireless network that uses land-based radio transmitters or base stations.

PDN (Public Data Network or Packet Data Network) - a network that provides packetized data services, such as the Internet.

GSN or xGSN (GPRS Support Node) - a generic term that refers to both SGSNs and GGSNs.

- SGSN (Serving GPRS Support Node) - sends and receives data from mobile stations, and maintains information about their location.
- GGSN (Gateway GPRS Support Node) - acts as mediator between encapsulated GTP traffic on the PLMN, and packetized IP traffic on the Internet and other PDNs.

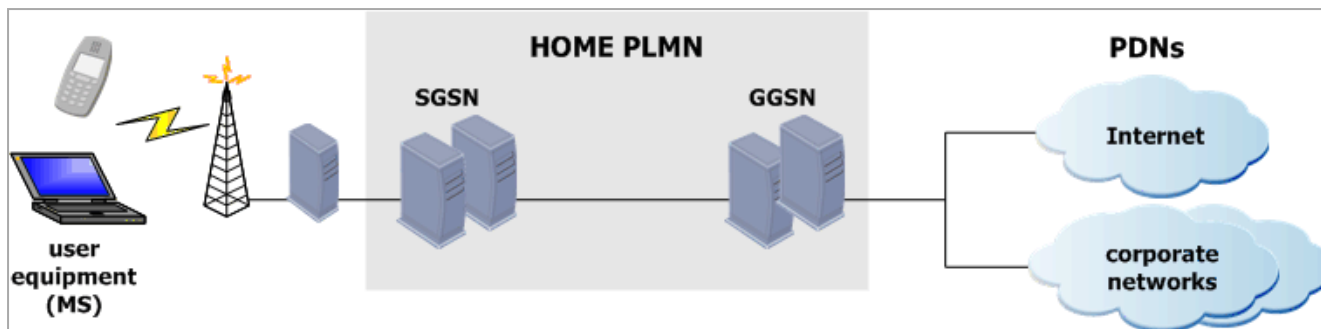
MS (Mobile Station) - a wireless device that uses a radio interface to access network services.

GRX (GPRS Roaming eXchange) - an IP network that connects PLMNs, enabling MSs to connect to their home PLMNs through roaming partners.

APN (Access Point Name) - provides routing information for SGSNs

PDF (Policy Decision Function) - logical element that uses standard IP mechanisms to implement policy in the IP media layer. The PDF uses policy rules to make decisions in regard to network based IP policy, and communicates these decisions to the PEP on the GGSN.

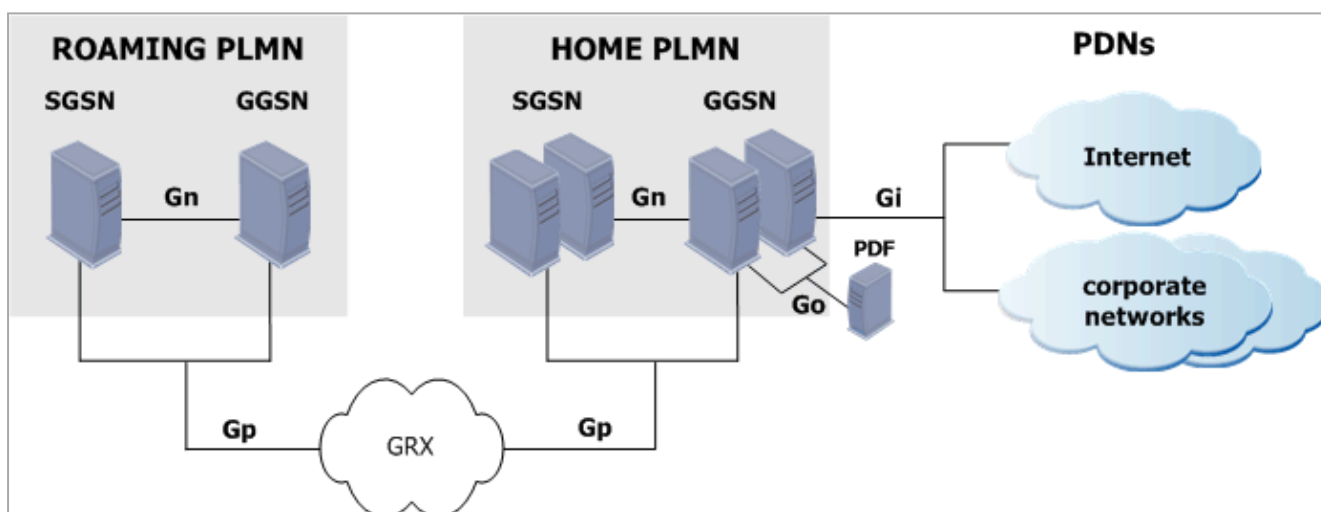
PEP (Policy Enforcement Point) -logical entity that enforces policy decisions made by the PDF. It resides on the GGSN.



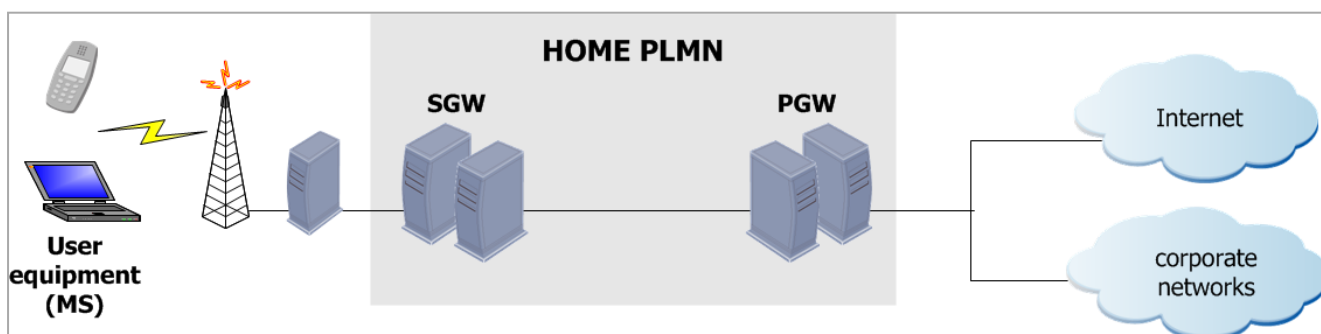
## Interfaces

An interface is the point of connection between telecommunication entities. While there are many types of interfaces in a cellular network, this guide deals primarily with these:

- Gi interface - connects GGSN to an external PDN.
- Gn interface - connects xGSNs on same PLMN.
- Go interface - connects a GGSN to a Policy Decision Function (PDF).
- Gp interface - connects xGSNs on different PLMNs.



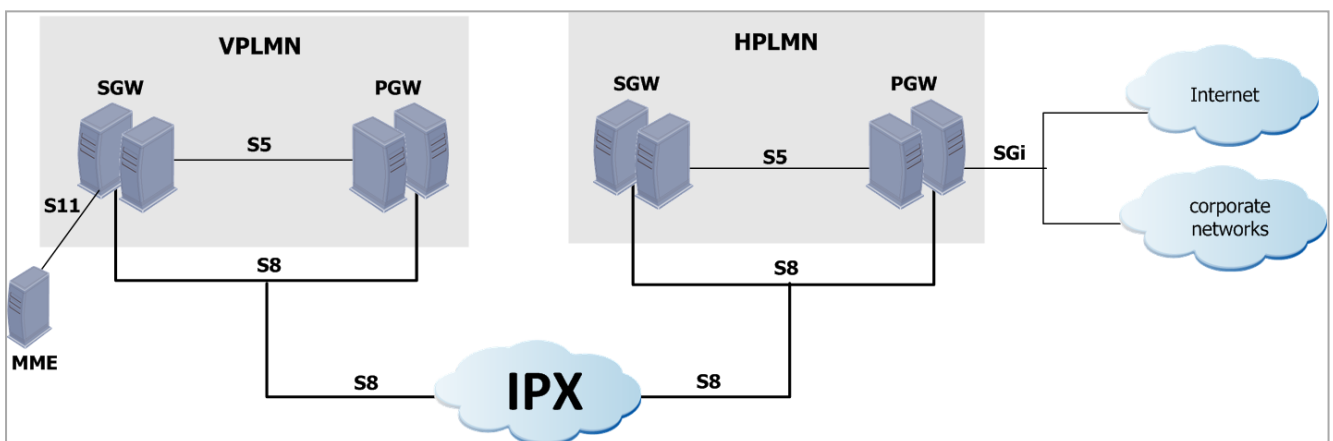
## Basic Components of LTE Networks



- SGW - The serving gateway, which handles routes and forwards data packets and eNodeB handovers.
- PGW - Packet Data Network Gateway, the exit and entry point for traffic to user equipment.
- MME - Mobility Management Entity, responsible for user equipment tracking and selecting the serving gateway for user equipment during the initial attach.
- HPLMN - Home Public Land Mobile Network, which identifies the PLMN (Public Land Mobile Network) that holds the subscribers profile.
- IPX - IPS Exchange - a model for the exchange of IP traffic between fixed and mobile operators, and other types of service providers such as ISPs.

## Interfaces

S5 and S8 are the main interfaces used for roaming. S5 is used in the Home Public Land Mobile Network (HPLMN) and S8 in the Visiting Public Land Mobile Network (VPLMN).



- SGi interface - connects PGW to an external PDN.
- S5 interface - connects SGW and PGW on same PLMN.
- S8 interface - connects SGW on the VPLMN with the PGW at the HPLMN.
- S11 interface - connects the MME to the SGW.

## Signaling Protocols

GTP (GPRS Tunneling Protocol) - used to transport user data between GSNs. The data is encapsulated inside a packet, which consists of the data payload and a routing header. GTP versions have been updated to include new capabilities, however most GPRS/IPX networks maintain support for both.

GTP-C (GPRS Tunneling Protocol - Control) - used for control messages to create, update and delete GTP tunnels, and for path management.

GTP-U (GPRS Tunneling Protocol - User) - used for user messages to carry user data packets, and signaling messages for path management and error indication.

TEID (Tunnel Endpoint Identifier) - used to unambiguously identify a tunnel endpoint.

G-PDU (GTP Protocol Data Unit) - used for data and control information.

PDP (Packet Data Protocol) - a network protocol used by an external packet data network (usually IP).

PDP address - the address of an MS in the external packet data network, also called End User IP address.

PDP context/session - a logical association between an MS and PDN. There are six types of PDP context commands:

- Create
- Update
- Delete
- Modify (only GTPv2)
- Request
- Response

For an extensive list of industry-specific terms, see the ["Glossary" on page 85](#).

## Comparing GTP Versions

The most important differences between GTP version 0 and version 1 arise from the fact that GTP version 1 supports several different services simultaneously, which in turn requires a clearer focus on Quality of Service (QoS). While the difference between GTPv0/v1 to GTPv2 is due to the change in the network nodes and their roles the main new node presented is the MME which in turn introduced new concept of tunneling different the main difference is that GTP tunnel is now addressed as session, with bearers that act as data link context.

## Port Changes

While the entire GTP version 0 communication is transmitted over a single UDP (3386), GTP version 1/2 packets are transmitted over two different UDP ports:

- The **Control plane**, which includes the create, update, delete, modify and echo exchanges, now uses UDP port 2123.
- The **User plane**, which includes the tunneled data packets, now uses UDP port 2152.

By separating signaling and mobile user traffic to two different ports, either one of these types of traffic can be encrypted without the other.

## Multiple PDP Contexts for the Same PDP Address

In GTP version 0, an MS might have several simultaneous PDP contexts, but a single PDP address on a specific APN is uniquely associated with a single PDP context. For each combination of external packet network and MS-local address, there is one tunnel (PDP context).

In GTP versions 1 and 2, multiple PDP contexts/sessions are allowed per PDP address and APN. After a successful GPRS activation, where the MS establishes a PDP context/session and connects to the external network, the MS can initiate more PDP contexts/sessions with the same APN.

The new PDP contexts/sessions for the same PDP address differ in QoS requirements and charging characteristics, and are used to separate streams of different services or protocols.

This is useful for IMS, where the initial PDP Context/sessions (used for SIP registration) have low bandwidth requirements, but the following PDP Contexts/sessions (used for actual data streaming) require a higher bandwidth.

# Introducing Carrier Security

Carrier Security was specifically designed for wireless operators and combines Check Point's patented Stateful Inspection technology with full GPRS Tunneling Protocol (GTP) awareness. Carrier Security inspects all GTP tunnel fields in the context of both the packet and the tunnel. This enables granular security policies that deliver the highest level of security for these wireless infrastructures.

## The Need for Security on GPRS/UMTS Networks

As networks undergo the transition from proprietary, closed networks to the open world of IP, both the MS and the network itself become exposed to the full range of security threats that plague the Internet, as well as to threats that specifically target wireless mobile networks. IP-based signals expose UMTS/LTE networks to a wide range of attacks: Denial of Service, IP spoofing, Overbilling, tunnel hijacking, flooding, DNS cache poisoning, and many other attacks. Any hacker with the will to do so can compromise security and disrupt communications.

## GTP - Insecure By Design

GTP, the key protocol used in delivering mobile data services, offers no solution. The GTP specification specifically states:

*No security is provided in GTP to protect the communication between different GPRS networks. The security is provided, if needed, between the Border Gateways in different GPRS networks by operator agreements. A security mechanism that may be considered is for example IP Security.*

The UMTS/LTE network is at risk from both its own subscribers and its partner networks. Despite years of development and deployment, new security vulnerabilities are constantly being found in IP software, and evolving UMTS/LTE software will be no different. The lesson: don't depend on the network to provide your security - provide your own security.

## Check Point Protects UMTS/LTE Networks

Check Point Carrier Security (formally known as Firewall-1 GX) has been protecting GPRS, UMTS and LTE (also known as 2.5, 3rd and 4th Generation) networks worldwide since 2001 by providing a GTP-level security solution that blocks illegitimate traffic "at the door." Only Carrier Security provides cellular operators, enterprises and end users with an integrated, open, end-to-end security solution. Carrier Security enables operators to define and enforce customized, granular "GTP-aware" security policies for all GTP protocol versions.

Carrier Security is fully configurable, and produces GTP-specific detailed logs and alerts. It can be installed on the:

- Gp/S8 interface (between the Home PLMN and the GRX network)
- Gn/S5 interface (between the nodes in the Home PLMN)

Working together with a standard Security Gateway on the Gi/SGi interface, Carrier Security provides UMTS/LTE networks with the highest level of security available today. Carrier Security provides protection from the following threats to the core network and mobile users:

Carrier Security protections for cellular networks:

| Attack Target | Attack Source                                                          | Deploy Carrier Security on:                       |
|---------------|------------------------------------------------------------------------|---------------------------------------------------|
| Core Network  | Partner network, or Partner of a Partner, or anyone with access to GRX | Gp/S8 interface                                   |
| Core Network  | Mobile Users                                                           | Gn/S5 interface                                   |
| Core Network  | Internet or enterprise connections                                     | Gn/S5 interface, with gateway on Gi/SGi interface |
| Core Network  | Within core network                                                    | Gn/S5 interface                                   |
| Mobile Users  | Mobile user to mobile user                                             | Gn/S5 interface                                   |

## The Check Point UMTS/LTE Commitment

Check Point maintains a dedicated R&D and product management team specializing in UMTS/LTE security. Check Point is fully committed to keeping its products up-to-date with UMTS and LTE standards.

## Logging, Alerts, and Reporting

In addition to security enforcement, Carrier Security provides a rich set of GTP-specific log information, including granular logging details on tunnel creation, updates and deletions. Administrators can set a wide range of security alerts based on defined activity. And Carrier Security provides a GTP-enhanced reporting tool to give a bird's-eye view of network activity. See chapter 5, "Monitoring GPRS Network Security" for more information on logging, alerts and reports.

# Licenses

All GTP features require carrier licenses installed on the Security Gateway. The management server does not require special licenses. If there is no Carrier license on a Security policy installation will fail.

Carrier license string: CPSG-CARR

## Before Installing Carrier Security

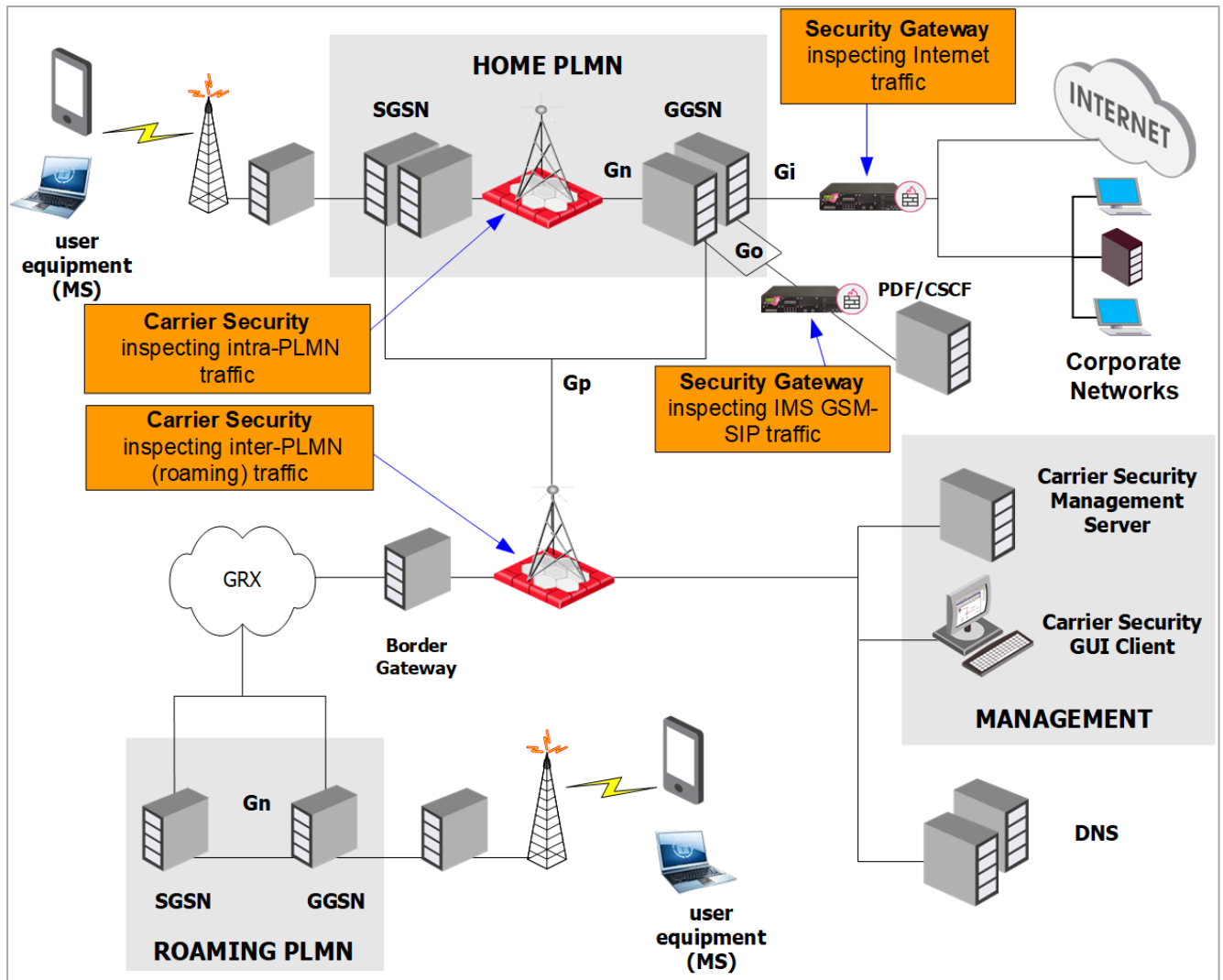
Before installing Carrier Security, familiarize yourself with the following Check Point documentation:

- Read the Release Notes and applicable Administration Guides on the R82 Home Page SK - sk181127.
- Read the R82 Known Limitations SK for the information concerning LTE features - sk181128.
- Read this Administration Guide, which describes the Carrier Security features.

## Deploying Carrier Security

For maximum security, a Carrier Security Gateway should be installed at all points in the network where the Home PLMN interfaces with other networks: at Border Gateways (Gp/S8) and Intra-PLMN interfaces (Gn/S5).

Suggested Carrier Security deployment:



In this example, two types of Check Point Gateways are deployed. The protections provided by each are described below:

### Carrier Security Gateways

Carrier Security Gateways (Gateways with carrier license) are deployed at these interfaces:

| Interface | Located Between   | Description                                                                                                                                                                                                                                              |
|-----------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gp/S8     | Home PLMN and GRX | Filters incoming roaming traffic and enforces a GTP-aware Security Policy, protecting the Home PLMN from malicious or erroneous traffic from the networks of roaming partners, as well as from traffic not originating from legitimate roaming partners. |

| Interface | Located Between                      | Description                                                                                      |
|-----------|--------------------------------------|--------------------------------------------------------------------------------------------------|
| Gn/S5     | GGSNs and the SSGNs in the Home PLMN | Filters traffic between the Home PLMN GSNs, protecting them from malicious or erroneous traffic. |

# Security Gateways

Security Gateways (Gateways without carrier license) can be deployed at these interfaces:

| Interface | Located Between          | Description                                                                                                                                                                                                                                   |
|-----------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gi/SGi    | Home PLMN and Internet   | Protects the network from threats originating from the Internet, such as the Overbilling attack.                                                                                                                                              |
| Go        | GGSNs and CSCF SIPserver | Protects the CSCF (Call Session Control Function) SIP server and enforces the SIP protocol. A major feature of this Gateway is RTP pinholing - i.e., it dynamically follows the negotiated RTP sessions, opening only the UDP ports required. |
| S11       | eNodeB, MME and SGW      | Inspects SCTP and Diameter performs IPSec encryption on GTP-U traffic.                                                                                                                                                                        |

**Note** - Mobile to mobile IMS communications can also be protected by the Gateway on the Go interface. To do so, mobile to mobile traffic must be routed from the GGSN to the Gateway and back to the GGSN.

# Securing UMTS/LTE Networks

Carrier Security includes a variety of measures to protect UMTS/LTE networks from possible attack. The GPRS Tunneling Protocol (GTP), the communications protocol of mobile networks, was designed without regard to security, and so any security scheme for GPRS/UMTS networks must account for the GTP protocol. Carrier Security thus enforces a security policy that is GTP-aware, capable of identifying and rejecting malicious data or attempts to misuse the protocol, and even inspect GTP-encapsulated data packets.

Additional security measures can be deployed at the Gn, Go, S1, S11 interface. By deploying a Security Gateway with advanced cellular network capabilities at the Gn and/or Go S1, s11 interface, the following additional security is available:

- at the Gn/S5 interface:
  - APN based security policy
- at the Go interface:
  - GSM-SIP aware firewall security over IPv4 or IPv6
  - SCTP and Diameter aware firewall security

This chapter presents the various protections that Carrier Security provides for UMTS/LTE networks, and is divided into the following sections:

- **GTP Protocol Security** covers how Carrier Security scans GTP communications for abuse of the protocol, and includes a summary of the Overbilling attack and the protection that Carrier Security provides.
- **GTP-Aware Security Policy** covers the principles of establishing a Security Policy that can selectively allow the various signaling messages within GTP, as well as the addresses from which the communications originate.
- **Intra-Tunnel Inspection** covers how Carrier Security inspects mobile user traffic encapsulated by GTP.
- **Configuring Security** explains how to create a basic Security Policy and configure the security features available with Carrier Security.

## GTP Protocol Security

Carrier Security inspects all GTP tunnel fields in the context of both the packet and the tunnel, which allows the definition and enforcement of a granular, GTP-specific Security Policy that protects both network assets and subscribers from possible attack. Carrier Security secures the GTP protocol in the following ways.

- **GTP protocol enforcement** ensures the legitimate use of the GTP protocol, protecting GSN servers from harmful traffic. The Carrier Security parser verifies that each GTP message contains the correct set of Information Elements (IE) in the proper sequence.
- **GTP Stateful Inspection** ensures that only legitimate GTP traffic passes through the gateway. For example, data packets (G-PDUs) and PDP update messages are allowed only for open PDP contexts/sessions. Carrier Security detects any tampering with the state and rejects such traffic.
- **PDP context/session timelines** are strictly verified, and operations on unestablished or deleted contexts are disallowed.

## Understanding the Overbilling Attack

The Overbilling attack targets a cellular operator's network, reputation, and bottom line. The attack takes advantage of a weakness in the architecture of GPRS networks where previously used IP addresses are reassigned to other devices. The attack takes place as follows:

1. Attacker connects to cellular network as a legitimate subscriber.
2. An SGSN/SGW assigns the mobile device an IP address.
3. Attacker uses a malicious server to continuously send packets to the address assigned in **step 2**.
4. Attacker ends the GPRS connection and the PDP context/session is deleted.
5. The malicious server ignores TCP FIN packets and continues to send packets to the address assigned in **step 2**.
6. An innocent subscriber requests service and the SGSN/SGW reassigns the original IP address.
7. The data traffic that the malicious server is sending to that address is charged to the new device's account, without the knowledge of its owner. The device ignores the traffic as noise, but its owner is charged for the time-slots occupied.
8. Attacker repeats the process, adding IP addresses to the attack each time a new one is assigned.

When invoices are distributed, the company is attacked again by irate customers who claim they didn't use this bandwidth. The net effect of this attack is inflated receivables estimates, a swamped customer/billing center, loss of customer loyalty, and great aggravation for the customer. The attacker in this way compromises the cellular provider's good name.

## The Check Point Solution to the Overbilling Attack

This attack was officially reported to the GSM Association in 2002, and the Carrier Security solution is the GSMA-approved ["Enabling Overbilling Attack Protection" on page 40](#) solution.

Briefly, Carrier Security protects GPRS networks from Overbilling attacks in this way:

- Carrier Security is deployed on the Gn/S5 interface, and a standard Security Gateway is on the Gi/SGi interface.
- Whenever a GTP tunnel is deactivated, Carrier Security notifies the Security Gateway on the Gi/SGi interface to block all packets belonging to connections of the de-activated tunnel. Thus the packets sent by the malicious server are stopped at the Security Gateway, and no further steps need to be taken.

## GTP-Aware Security Policy

Carrier Security provides the ability to define a single GTP-aware Security Policy using Check Point's intuitive GUI tools. The next section describes how you can use GSN address filtering, GTP Tunnel, Path, and Mobility Management signaling messages to create a GPRS Security Policy.

### GSN Address Filtering

The ability of Carrier Security to identify the various aspects of GTP traffic allows for the enforcement of security rules in a granular manner. One such capability is the ability to enforce the creation and update of PDP contexts/sessions by defined GSN addresses.

- **PDP context/session creation** is enforced according to directional security rules that identify the range of SGSN/SGW addresses that are allowed to create tunnels.
- **PDP context/session updates, redirection and handover** are enforced according to directional security rules. In addition, Carrier Security strictly enforces SGSN/SGW handovers and GSN redirections according to predefined address ranges and sets (Handover Groups).

### GTP Message Type Filtering

GTP Message Type Filtering refers to the gateway's ability to allow only those message types categorized by an interface profile. Each interface profile represents a list of message types to enforce (accept or drop actions).

You can select an interface profile for a particular service. For example if you select an interface profile named "S8" that includes message types: 1, 2, 3, 4, 5 and then enforce a DROP action for this service, all of the message types are dropped.

There is also an option to set a "Custom" message type field. Use this field to manually input message types that you want to accept or drop.

Note that:

- GTP Message Type Filtering supports GTPv1 and GTPv2 only.
- "Any" is the default option - when selected, the relevant service matches any message type.

- The Gateway enforces control message types after it establishes a GTP tunnel. If according to the service an incoming message type does not have permission to perform actions on the GTP tunnel, the gateway does not match the packet.

## GTP Tunnel Management / User Traffic

Carrier Security recognizes and can control the access of GTP Tunnel management services to your network. You can build simple security rules to allow all GTP user traffic on your network, or opt for a more refined policy through the various customizations possible with Carrier Security. The following sections discuss the various ways you can secure user traffic on your UMTS/LTE network.

## Tunnel Management and User Traffic Service

Tunnel management services are those parts of the GTP protocol that carry the actual user traffic - voice, data, for example. Carrier Security allows you to build security rules to allow this traffic within your network. Use these services to enable communication between incoming traffic and your GSNs. Tunnel management services on Carrier Security are predefined as:

- **gtp\_v0\_default**, which addresses message types defined in 3GPP TS 09.60.
- **gtp\_v1\_default**, which addresses message types defined in 3GPP TS 29.060.
- **gtp\_v2\_default**, which addresses message types defined in 3GPP TS 29.274.

For more information about using tunnel management services in security rules, see: ["Creating Security Rules with GTP Services" on page 37](#)

## GTP Service Filters

You can be more selective than simply allowing all tunnel traffic, however. You can build security rules to more precisely select which user traffic can cross the firewall. Through the use of GTP-aware filters, Carrier Security provides the ability to limit the creation of PDP contexts/sessions to traffic that matches specific parameters that you define. You can set up a GTP service filter that matches traffic by:

- APN
- IMSI (MCC, MNC) Prefix
- MS-ISDN Prefix
- APN Selection Mode
- LDAP Group
- Radio Access Technology
- GTP Message Type

As cellular operators tend to sort their LDAP databases by either IMSI or MS-ISDN, Carrier Security can identify whether a user belongs to a specific LDAP group by IMSI or MS-ISDN prefix.

By customizing the pre-defined user traffic services **gtp\_v0\_default**, **gtp\_v1\_default**, **gtp\_v2\_default**, or creating new customized services, you can build a logical "AND" argument to choose what specific characteristics to match, and then configure a security rule to accept this specific class of user traffic. While predefined GTP services are provided with Carrier Security, it is recommended that you create new services for customization.

For configuration information, see: ["Customizing GTP Services" on page 45](#)

## End User PDU Protections

Carrier Security protects the integrity of Protocol Data Units (PDUs).

- **Sequence Number Validation** - Carrier Security verifies PDU sequence numbers for both signaling and user traffic. For configuration information, see ["GTP PDU Integrity Tests" on page 50](#)
- **Flow Labels Validation** - In GTP ver. 0, when two GTP tunnels are open on one device, they have the same tunnel ID. To distinguish between tunnels, Carrier Security adds packet flow labels to the tunnel ID. To secure this solution, Carrier Security can validate that the flow labels assigned belong to packets of a similar flow.
- **Multiple GGSN/PGW Interface Support** - GTP ver. 1/2 allows xGSNs to reply to PDP context/sessions requests from a different interface than the one to which the request was sent. This capability, useful for load balancing, can make a system vulnerable to Session Hijacking. Carrier Security is able to protect against Session Hijacking through the use of Handover Groups. For configuration information, see ["Secure Connectivity Features" on page 51](#)

## End User Address Modes

Carrier Security can be configured to block PDP context creations from MSs with static IP addresses.

For configuration information, see **Allow usage of static IP address** in ["Secure Connectivity Features" on page 51](#)

## IPv6 T-PDU Security

Carrier Security enforces the proper use of IPv6 T-PDUs encapsulated inside IPv4 GTP G-PDUs. The IPv6 End User Address Information Element in the Create PDP Context/session message is inspected and allowed upon proper validation. The IPv6 End User address in the T-PDU is then tested for GTP Anti-Spoofing. For more information, see: ["GTP Address Anti-Spoofing" on page 31](#)

Carrier Security also supports End User Address Mode IPv4v6 that can hold two IP address types at the same time (one IPv4 and one IPv6).

## Session Hijacking Protection through GSN Handover Groups

When an SGSN/SGW sends data to a GGSN/PGW and awaits a reply, an unprotected GPRS/UMTS network may become exposed to what is known as Session Hijacking. Session Hijacking is a type of attack that involves eavesdropping on an established communications session and assuming the identity of an authenticated user. It can occur during handover, redirection, and by permitting "unknown" GGSNs/PGWs to reply to SGSN/SGW-initiated communications.

Handover, which enables subscribers to seamlessly move from one area to another with no interruption of active sessions, is a fundamental feature of UMTS/LTE. But this ability to move from SGSN/SGW to SGSN/SGW can expose the cellular operator to Session Hijacking. In GTP, a GGSN/PGW will relinquish control of a tunnel to any device from which it receives a PDP context/session update request message. This can be exploited to hijack GTP tunnels or cause a Denial of Service (DoS).

Redirection, which enables load sharing among xGSNs, is also vulnerable to Session Hijacking. In some GTP signaling messages, a malicious xGSN can redirect the handling of subsequent messages to another device by inserting that device's IP address in the message.

A cellular operator may choose to set up multiple GGSNs, and the GTP protocol allows a GGSN/PGW other than the one that received an SGSN/SGW message to reply to that message. This practice can leave a network vulnerable to Session Hijacking, where a malicious GGSN/PGW responds to an SGSN/SGW message before the real GGSN/PGW can respond. Because the SGSN/SGW has been configured to allow any response, it directs traffic to the malicious GGSN/PGW.

### Check Point's Solution - Handover Groups

To protect against this threat, Carrier Security uses Handover Groups, sets of IP addresses among which handovers are allowed. Handover Groups typically consist of the IP addresses of a UMTS/LTE operator's GSNs.

Carrier Security strictly enforces Handover Groups, allowing handover and redirection only within the group. In addition, tunneled and signaling packets are matched according to tunnel ID against a tunnel internal state table. Improper use of GTP signaling is blocked and logged. For configuration information, see ["Creating a GSN Handover Group" on page 36](#)

## Deactivating Session Hijacking Protection

The Session Hijacking protection is turned on by default. You can deactivate it with one command.

To deactivate the Session Hijacking protection, run this command on the Security Gateway in the Expert mode:

```
fw ctl set int gtp_allow_ho_bypass 1
```

To activate the Session Hijacking protection again, run this command on the Security Gateway in the Expert mode:

```
fw ctl set int gtp_allow_ho_bypass 0
```

To permanently disable the Session Hijacking protection, add this line to the `$FWDIR/boot/modules/fwkernel.conf` file and reboot:

```
gtp_allow_ho_bypass=1
```

## GTP Path Management Message Support

According to the protocol specification 3GPP, a path is a UDP/IP connection used to multiplex GTP tunnels between GPRS network resources, such as from one xGSN to another. The GTP signaling messages that maintain the GPRS tunnels are known as Path Management. Carrier Security allows you to build security rules to allow this traffic within your GPRS network. Use these services to enable communication among your xGSNs.

Path management services on Carrier Security are predefined as:

- **gtp\_v0\_path\_mgmt**, which addresses message types defined in 3GPP TS 09.60.
- **gtp\_v1\_path\_mgmt**, which addresses message types defined in 3GPP TS 29.060.
- **gtp\_v2\_path\_mgmt**, which addresses message types defined in 3GPP TS 29.274.

For more information about using path management services in security rules, see [x"Creating Security Rules with GTP Services" on page 37](#)

Use these Carrier Security features to protect your network from various attempts to abuse Path Management signaling messages:

- **GTP Echo Stateful Inspection** - ensures that there is a matching echo request in the log before allowing an echo response packet through the Security Gateway.
- **Limit Echo Rate** - The GTP protocol states that "an echo request shall not be sent more often than every 60 seconds per path." If desired, you can enforce echo requests as the protocol specifies, or to whatever interval you want. Carrier Security can be configured to restrict the echo rate per GTP path or allow GTP echo exchange only between GSNS with an active PDP context.
- **Limit Echo Retransmissions**- ensure that more than a defined amount of echo messages will not be sent on the same path.
- **GTP signal packet rate limit enforcement** -can be configured to limit the rate of signaling PDU to prevent signaling flooding or Denial of Service (DoS) attacks.
- **GTP HO packets rate limit enforcement** - can be configured to limit the rate of signaling packets from handover group to prevent flooding or Denial of Service (DoS) attacks.

For configuration information of Path Management, see: ["Limiting Signaling Rates" on page 52](#)

## GTP Mobility Management Message Support

Mobility Management messages are the control plane messages as defined in 3GPP TS 23.060 and 3GPP TS 24.008. They are sent between SGSNs during the GPRS Attach and Inter SGSN Routing Update procedures in GTPv1 while in GTPv2 they are handled by the MME. Carrier Security enforces protocol integrity and security policy for mobility management messages. Carrier Security also allows you to isolate these messages within GTP packets and permit them access to your network nodes, while denying access to other types of GTP traffic. This capability can be used to enhance network security by allowing only mobility management GTP message traffic between xGSNs.

Carrier Security allows you to build security rules to allow this traffic within your GPRS network. Use these services to enable mobility management communications among your xGSNs. Mobility Management services on Carrier Security are predefined as:

- **gtp\_mm\_v0\_default**, which addresses message types defined in 3GPP TS 09.60
- **gtp\_mm\_v1\_default**, which addresses message types defined in 3GPP TS 29.060
- **gtp\_mm\_v2\_default**, which addresses message types defined in 3GPP TS 29.274

See ["Creating Security Rules with GTP Services" on page 37](#)

For more information about using path management services in security rules, see ["Creating Security Rules with GTP Services" on page 37](#).

## GTP MS Info Change Reporting Message Support

The Location Change Reporting messages defined here are control plane messages used in accordance with 3GPP TS 23.060 [4], section 15.1.1a, and 3GPP TS 23.203 [39].

**Note** - The 3GPP defines this service only on the GTPv1 protocol.

MS Info Change Reporting service does not exist by default in SmartConsole. It is necessary to manually define it.

**To define the MS Info Change Reporting service in SmartConsole:**

1. From the **Services** tree, right-click **Other > New Other**.

The Other Services Properties window opens.

2. In **Name**, enter:

`gtp_v1_ms_info`

3. In **IP Protocol**, enter:

`17`

4. Click **Advanced**.

The Advanced Other Service Properties window opens.

- 5. In the **Match** field, copy this text:

```
(dport = GTP_C_V1_PORT,GTP_MSGTYPE = 0x80) or (sport = GTP_C_V1_PORT,GTP_MSGTYPE = 0x81),gtp_get_ver(sr8, 3),sr8 = 1,set r_mflags (r_mflags | MFLAGS_UDP_REPLY),set r_mhandler &gtp_code,((set sr3 0, get <conn, 5> from gtp_paths to sr3) or record <conn,5; r_crule, 0> in gtp_paths)
```

- 6. Select **Accept Replies**.
- 7. Keep the defaults of other options.
- 8. Click **OK**.

The Other Service Properties window opens.

- 9. Click **OK**.

The gtp\_v1\_ms\_info service is now defined and can be used in the Rule Base. By default, the gateway applies stateful inspection on MS Info messages. Stateful inspection can be disabled by setting a kernel variable.

**To Enable or Disable Stateful Inspection on MS Info Messages:**

- 1. On the Carrier Security Gateway, edit this file:

```
$FWDIR/boot/modules/fwkernel.conf
```

- 2. If the file does not exist, create it:

```
touch $FWDIR/boot/modules/fwkernel.conf
```

- 3. Add one of these lines:

| Line                        | Meaning                                         |
|-----------------------------|-------------------------------------------------|
| gtp_enforce_ms_info_state=0 | Disable stateful inspection on MS Info messages |
| gtp_enforce_ms_info_state=1 | Enable stateful inspection on MS Info messages  |

- 4. Save the changes in the file and exit the editor.
- 5. Reboot the Security Gateway.

For more details about kernel parameters, see [sk26202](#).

## Dynamic Configuration of New GTP Messages and Information Elements

Carrier Security is aware of all commercially used 3GPP Release-13.3.0 GTP messages and Information Elements. It is however possible that new GTP messages (from 3GPP Release-13.3.0 and higher) and new GTP Information Elements will be used by GSN equipment in the future, and that these messages and Information Elements will not be known to Carrier Security "out of the box". Another scenario is when GSN equipment is using different release versions than the version Carrier Security supports. To expedite support for these innovations/mismatches, new GTP messages and Information Elements can be defined on the Carrier Security management station.

## Intra-Tunnel Inspection

One of the fundamental features of GTP is to encapsulate underlying (also known as *end user* or *subscriber*) protocols within the UMTS/LTE backbone network. User data is *tunneled* between GSNs, which means the data payload is encapsulated inside a GTP packet. Carrier Security can inspect the GTP traffic and enforce a Security Policy based on the encapsulated protocols. The following sections deal with the ability of Carrier Security to secure the GPRS network from malicious tunneled data.

### GTP Address Anti-Spoofing

Spoofing is the act of impersonating an end user IP address, usually for malicious purposes. Carrier Security enforces the proper use of end user IP addresses for IPv4 and IPv6 end user headers in tunneled GTP packets (G-PDUs). During tunnel establishment, the end user's IP address is exchanged. Carrier Security verifies that all G-PDUs that are exchanged using this tunnel use a matching IP address as the user side IP address.

During PDP context/session establishment, an end user IP address is exchanged. This address is stored in the state information table of the tunnel, and is used by the MS in the subsequent G-PDUs. In the GTP dynamic mode, this IP address is allocated by the GGSN/PGW and is sent back in the create PDP context/session reply. In the static mode the subscriber has a constant (static) IP address. In this case the MS sends this IP address to the SGSN/SGW, who in turn sends it to the GGSN/PGW in the create PDP context/session request.

Carrier Security can be configured to inspect each G-PDU on a specific tunnel for malicious use of the end user IP address. For configuration information, see ["GTP Intra Tunnel Inspection and Enforcement" on page 49](#).

### GTP Address Anti-Spoofing for IPv4

In IPv4, the end user IP address in the tunneled IP packet's header is compared to the tunnel's established end user IP address. If the addresses are different, the packet is dropped and logged.

## GTP Address Anti-Spoofing for IPv6

In IPv6, the end user IP address in the tunneled IP packet's header is compared to the tunnel's established end user IP address. Carrier Security allows the following IPv6 address scenarios:

- The prefix of the IPv6 address equals the tunnel's established end user IPv6 prefix.
- If the IPv6 address is a Link local address and its identifier equals the tunnel's established end user IPv6 identifier.

In addition, Carrier Security allows the following IPv6 address types.

- An unspecified address :: (0::0) in T-PDUs originating from an SGSN.
- A Multicast IPv6 address in T-PDUs originating from a GGSN.

If an IPv6 address appears in any other form, the packet is dropped and logged.

## Block GTP in GTP

An SGSN converts mobile user traffic to encapsulated GTP when passing the data to a GGSN. This encapsulation can be exploited by an attacker, whereby a mobile user injects a forged GTP packet, which is in turn encapsulated by the SGSN. This packet could contain an instruction to the GGSN, such as to disconnect users. Carrier Security can detect and block GTP encapsulated inside GTP.

For configuration information, see: ["GTP Intra Tunnel Inspection and Enforcement" on page 49](#).

## MS to Gn/S5 Network Policy Enforcement

If the IP addresses of the servers on an unprotected Gn/S5 network were to become known to a potential attacker, a Mobile User could exploit the fact that the GGSN/PGW will route back to the Gn/S5 network T-PDU packets with a destination address of the Gn/S5 network, and initiate a Telnet session or other forms of unauthorized communications to attack the system. While some GGSNs/PGWs are capable of blocking this type of attack, others are not.

Carrier Security can be configured to deny any attempt by a Mobile User to send data to the Gn/S5 network. This is an extension of the Block GTP in GTP feature, i.e., disabling additional potential attacks.

MS to Gn network traffic is blocked by creating an APN object to represent the Gn/S5 network and then blocking all user traffic from IP addresses belonging to real APNs destined for the Gn/S5 network. For configuration instructions, see ["Blocking MS to Gn/S5 Network Traffic" on page 48](#)

## APN Domain End User Address Enforcement

An Access Point Name (APN) provides routing information for SGSNs/SGWs and GGSNs/PGWs. The APN, which is written as a string, contains the identity of the external service requested (the Operator ID) by the MS, and routing information (the Network ID). The two IDs are written something like this:

Operator ID: `MNC1234.MCC5678.gprs`

Network ID: `example.net`

Check Point has taken APNs a step further, integrating support of domains with APNs. A domain, consisting of addresses, IP subnets, address ranges or groups thereof, may be configured on an APN object. APN Domains specify the range of IP addresses that are assigned to MSs upon connecting to an APN. For example, you can create one APN called *Content\_Servers* that assigns a range of IP addresses from 10.1.1.1 to 10.1.10.255, and another called *Internet*, that assigns from a range of 192.168.1.1 to 192.168.10.255.

You can also use APN objects to define rules that specify things like: from which networks PDP contexts/sessions for enterprise APNs may be created, or to grant the CEO sole access to a specific APN, or to accept Handovers only between specified networks.

When a PDP context/session is created in which the exchanged end user IP address does not belong to the configured domain, the context will be dropped and logged.

## Wildcard APN Matching

To further enhance the ability to define and include certain APN addresses in security rules, Carrier Security allows the use of wildcards in APN matching. For example, you can create an APN object called `*.example.gprs` that will match APN object strings like `123.example.gprs` and `abc.example.gprs`.

For configuration information, see:

- ["Creating an APN Object" on page 47](#)
- ["GTP Intra Tunnel Inspection and Enforcement" on page 49](#)

## MS to MS Policy Enforcement

Carrier Security can be configured to prevent undesirable traffic between two end users (MSs) simultaneously connected to a PLMN. There are two variations of this capability: the ability to block intra-tunnel traffic between MSs of the same APN, and the ability to block user plane traffic between MSs of different APNs.

It is possible to enforce the correct use of *server side IP addresses* in tunneled GTP packets (G-PDU). Server side IP addresses refer to the IP address in the G-PDU header not belonging to the mobile subscriber, but to the server (host) with which the MS is communicating. For G-PDUs traveling from the SGSN/SGW to the GGSN/PGW, the destination IP address of the G-PDU is considered to be the server side address. For G-PDUs traveling from the GGSN/PGW to the SGSN/SGW, the source IP address of the G-PDU is the server side address.

Each G-PDU is inspected for malicious use of server side IP address. The server side IP address in the tunneled IP packet's header is compared to the relevant predefined APN address domains, and if the address is found to be in one of those disallowed domains for this tunnel, then the packet is dropped and logged.

Note the following:

- MSs that are connected using tunnels of APNs that are configured to block non-desirable MS to MS traffic are protected.
- APN domains that are searched for possible violation of the inter-APN enforcement are global (all defined APN domains, except the one in whose context we are currently inspecting), and therefore they are not dependent on the current APN context.
- Only local APNs need to be defined in the system for the purpose of this feature. This feature does not require configuration of roaming providers' APNs. The reason for this is that packets of PDP contexts belonging to roaming operators' APNs should never connect to the local GGSN.
- Configuration of only local APNs will not interfere with visiting MS traffic since GTP tunnels used by such users belong to external operator APNs.

For more information on configuring APN objects, see: ["GTP Intra Tunnel Inspection and Enforcement" on page 49](#).

## Mobile Subscriber Traffic Security

Mobile Subscriber Traffic Security, sometimes referred to as *Full Intra Tunnel Inspection*, enables real Security Gateway filtering for T-PDU traffic (i.e., decapsulated G-PDU). T-PDU traffic can now be processed by the Security Gateway inspection engine. This protection is selectable per GTP service, and effectively enables the following security measures for Mobile User traffic:

- Security Gateway Policy, including optional Anti-Spoofing protection.
- IPS protections.
- Application protections.
- Event Logging and Reporting. An IMSI field, which indicates which mobile user is linked to a logged event, can be added to every log generated, thereby eliminating reliance on End User IP address for identification.

This feature works by first passing the G-PDU through the regular GTP engine. If the G-PDU matches a GTP service where Mobile Security Traffic filtering has been enabled, the G-PDU is decapsulated and then analyzed with the security measures listed above. If the decapsulated packet is dropped, the outer packet will be dropped as well.

Mobile Subscriber Traffic Security can be enforced per GTP service. This means that a Security Policy can be implemented that inspects traffic from certain partners, and not from others. Intra Tunnel Inspection is fully supported, however, only with IPv4 in environments with unfragmented external (G-PDU) and internal (T-PDU) packets, and without overlapping IP addresses.

This is a very powerful feature - enabling true and full intra tunnel inspection for user traffic at the Gn, Gp or S5/S8 interfaces. To enable these protections on GTP services, see ["Customizing GTP Services" on page 45](#).

## Configuring Security

This configuration information refers only to the cellular network features provided by Carrier Security.

For information about configuring other aspects of Check Point software, refer to the applicable documentation.

You should have:

1. Installed:
  - Management Server
  - SmartConsole GUI
  - Carrier Security Gateway
2. Opened SmartConsole and connected to the Management Server.

The initial configuration of Carrier Security involves:

- Creating GSN Objects
- Creating a GSN Handover Group
- Setting up a Security Policy

Follow these steps to set up basic security. To customize your security policy, see: ["Enforcing a More Granular GTP Security Policy" on page 45](#).

## Creating GSN Objects

Use SmartConsole to create objects representing your GSNs and the GSNs of your roaming partners.

**Note** - To create GSN objects for your roaming partners, you should obtain a list of their IP addresses.

1. For the Home PLMN, define a Host object for an xGSN.
2. In the **Network Objects** tree, right click on **Nodes**, and select **New > Host**.
3. Enter an identifying name and the xGSN's IP address.
4. Repeat for each SGSN/SGW and GGSN/PGW on your network.
5. For the roaming partners, define a Host object for an xGSN.
  - If the GSN has a single IP address, right click on **Nodes**, and select **New > Host**. Enter an identifying name and the xGSN's IP address. Repeat for each roaming partner SGSN/SGW and GGSN/PGW.
  - If the GSN has a range of IP addresses or subnets, right click on **Network Objects**, and select **New > Address Range**. Enter an identifying name and define subnets or IP address ranges to represent the packets coming from or intended for the GSN. Repeat for each roaming partner SGSN/SGW and GGSN/PGW.

## Creating a GSN Handover Group

GSN Handover Groups prevent Session Hijacking and enable secure handover, redirection, and multiple GGSN interface support on Carrier Security. You should create Handover Groups for your own GSNs, and for your roaming partners' GSNs. Typically there will be one GGSN/PGW Handover Group and one SGSN/SGW Handover Group for each operator.

After you define your Handover Groups, include them as source and destination addresses in the rule base in order to allow handover, redirection, and multiple GGSN/PGW interface support for the GSNs that you included in the Handover Groups.

To define a GSN Handover Group:

1. In the **Network Objects** tree, right click on **Groups**, and select **New Groups > GSN Handover Group**.
2. Give the group a name, such as `SG_Home_HG`, where `SG` stands for SGSN/SGW.
3. Add the appropriate GSN objects you created in **Creating GSN Objects**.
4. To rate limit signaling packet from the GSN handover group select **Enforce GTP signal packet rate limit from this group** and enter an integer value (in PDU/sec).

To simplify the Security Policy rule base, it is recommended that you:

- Define a group consisting of your entire home PLMN GSNs.
- For each of your partner networks, define two groups:
  - one consisting of all of its SGSNs/SGWs or IP address ranges as appropriate
  - one consisting of all of its GGSNs/PGWs or IP address ranges as appropriate

## Configuring GSN Handover Group Limits

You can specify tunnel limits, for GTP handover groups. A newly created tunnel counts against the limit for handover groups on both sides of the tunnel.

To configure GSN handover group limits:

1. Connect with [Database Tool \(GuiDBEdit Tool\)](#) to the Management Server.
2. Search for **gtp\_groups\_limit\_enabled** and set the value to **true**.

Search for **gtp\_group\_percentage\_limit** and set the value to an integer that defines the percentage (0 - 100) of the tunnel capacity assigned to all handover groups.

The default value is 0 (unlimited).

This percentage applies to all groups for which no limit is defined explicitly.

3. Search for **gtp\_tunnels\_limit**.

This parameter is the maximum number of GTP tunnels that a handover group can create. Set this limit to make sure that one group does not take too much of the GTP tunnel allowance.

4. Search for **gtp\_tunnel\_group\_limit** in each handover group.

This value is an integer that defines the maximum number of tunnels that can be open for the specified group. The default value is 0 (not defined).

The explicitly defined **gtp\_tunnel\_group\_limit** has precedence over the **gtp\_groups\_percentage\_limit** definition.

## Creating Security Rules with GTP Services

Allow GTP traffic on your network by creating rules using GTP services. There are six predefined GTP related services in Carrier Security:

- Path management - **gtp\_v0\_path\_mgmt** and **gtp\_v1\_path\_mgmt** (User Defined services) or V2.
- Tunnel management - **gtp\_v0\_default**, **gtp\_v1\_default** and **gtp\_v2\_default** (Tunnel services and User traffic).

You can use these predefined defaults or create a custom GTP service.

- Mobility management - **gtp\_mm\_v0\_default** and **gtp\_mm\_v1\_default** (Mobile User services).

## Creating a new GT Tunnel Management Service

This service does not include path management

### To create a new GTPv2 Service

1. In **SmartConsole**, open the **Object Explorer**.
2. Click > **New** > **Service** > **GTP**.
3. Enter a name for the new service.
4. On the **General** page, select a GTP version: **V0**, **V1**, or **V2**.
5. If you selected the **V2 Additional** service, select one or more service types:
  - **Trace Management**
  - **CS Fallback and SRVCC**
  - **Restoration and Recovery**

### For the Home PLMN

1. Create a Tunnel Management rule to restrict tunnels and user traffic on your network to your GGSNs from only your SGSNs.
2. Create a Path Management rule to enable path management services between SGSNs and GGSNs.

**Note** - Use the GSN Handover Groups you created in **Creating a GSN Handover Group** as the Source and Destination objects in the security rules.

The next table depicts a basic security policy consisting of these two rules:

### Basic Security Policy

| Source                   | Destination              | VPN | Services & Applications                                  | Action | Track | Install On     | Comment                          |
|--------------------------|--------------------------|-----|----------------------------------------------------------|--------|-------|----------------|----------------------------------|
| SG_Home_HG               | GG_Home_HG               | Any | gtp_v0_default<br>gtp_v1_default<br>gtp_v2_default       | Accept | Log   | Policy Targets | Tunnel Management & User Traffic |
| SG_Home_HG<br>GG_Home_HG | GG_Home_HG<br>SG_Home_HG | Any | gtp_v0_path_mgmt<br>gtp_v1_path_mgmt<br>gtp_v2_path_mgmt | Accept | Log   | Policy Targets | Path Management                  |

To enable Mobility Management between SGSNs, the rule should look something like this:

## Mobility Management Rule

| Source     | Destination | VPN | Services & Applications                                     | Action | Track | Install On     | Comment                                 |
|------------|-------------|-----|-------------------------------------------------------------|--------|-------|----------------|-----------------------------------------|
| SG_Home_HG | SG_Home_HG  | Any | gtp_mm_v0_default<br>gtp_mm_v1_default<br>gtp_mm_v2_default | Accept | Log   | Policy Targets | Allow Mobility Management between SGSNs |

## For Roaming Partners

1. Add an inbound roaming traffic Tunnel Management rule to allow mobile subscribers to my network to connect from partners' SGSNs to my GGSNs.
2. Add an outbound roaming traffic Tunnel Management rule to allow mobile subscribers of my roaming partners' networks to connect to their GGSNs through my local SGSNs.
3. Add a Path Management rule to enable those services over both networks.
4. Add a reverse rule to accept PDUs from the GGSN to the SGSN on a previously established PDP context even if these PDUs are sent over ports that do not match the ports of the established PDP context.

Roaming partner security rules should look something like this:

**Roaming Partner Rules:**

| Source                                  | Destination                             | VPN | Services & Applications                                  | Action | Track | Install On     | Comment                                    |
|-----------------------------------------|-----------------------------------------|-----|----------------------------------------------------------|--------|-------|----------------|--------------------------------------------|
| PartnerA_HG                             | GG_Home_HG                              | Any | gtp_v0_default<br>gtp_v1_default<br>gtp_v2_default       | Accept | Log   | Policy Targets | Roaming for my users on partner's networks |
| SG_Home_HG                              | PartnerA_HG                             | Any | gtp_v0_default<br>gtp_v1_default<br>gtp_v2_default       | Accept | Log   | Policy Targets | Roaming for partners' users on my network  |
| PartnerA_HG<br>SG_Home_HG<br>GG_Home_HG | PartnerA_HG<br>GG_Home_HG<br>SG_Home_HG | Any | gtp_v0_path_mgmt<br>gtp_v1_path_mgmt<br>gtp_v2_path_mgmt | Accept | Log   | Policy Targets | Path management across networks            |

**Notes:**

- Under **Service**, specify either the GTP service, as appropriate to the partner GSN.
- In rules with a GTP service, the Reject action rejects the connection and sends the subscriber a "User Not Authenticated" PDU.
- GTP-U messages cannot match GTPv2 services in Firewall rules. You must also include the GTPv1 service in the rule to match GTP-U messages.

5. Install the Security Policy on the Carrier Security Gateways.

To further refine your Security Policy, see: ["Enforcing a More Granular GTP Security Policy" on page 45](#)

## Enabling Overbilling Attack Protection

Protection against Overbilling attacks can be implemented quickly and simply on networks with both a Carrier Security Gateway and a Security Gateway on the Gi interface.

Carrier Security resolves the vulnerability inherent in the GTP protocol by sending a `clean state` message to the Security Gateway whenever a PDP context is deleted. Configuration of this feature differs between systems where the Gi/SGi and Carrier Security Gateways are managed by the same management server, and those managed by different management servers.

## If the SGi and Carrier Security Gateways are managed by One Management Server

To protect your UMTS/LTE network from an Overbilling attack, do the following:

1. Create an Overbilling Group object:
  - a. From the **Network Object** tree, right click on **Group**, then select **New Groups > Simple Group**.
  - b. Name the group (e.g., `overb_gw_group`).
  - c. Select and add to the group the gateways that will receive the `clean state` message whenever a PDP context is deleted.
  - d. Click **OK**.
2. Enable Overbilling attack protection:
  - a. From the **Network Object** tree, double click the Gateway object of a Carrier Security Gateway.
  - b. Select the **Carrier Security** tab.
  - c. Check the **Send "clean state" request on each GTP tunnel Deactivation** property.
  - d. Select as the **Target** the group you created in **step 1**.
  - e. Repeat for each Carrier Security Gateway.

Define a rule allowing `FW1_sam` traffic from the Carrier Security cluster IP address to the Gi/SGi Gateways/Members.

| Source                   | Destination      | Service | Action |
|--------------------------|------------------|---------|--------|
| Carrier Security Gateway | Security Gateway | FW1_sam | Accept |

3. Install the security policy on each Security Gateway.

## If the Gi/SGi and Carrier Security Gateways are managed by Different Management Servers

Enabling Overbilling protection in environments with multiple management servers is a more complicated procedure. You must make modifications to the management and gateways for both Security Gateways and Carrier Security, including:

- Add a rule to the rule base that allows SAM traffic from the Carrier Security Gateway to the Security Gateway.
- Set SAM traffic to use SSL.

- Establish a trust relationship between the gateways.
- Follow the steps below precisely, and then test the solution according the instructions in *Testing Overbilling Protection*.

## On the Carrier Security Management Server

On the Management Server, use SmartConsole to define each Gi Member as an Externally Managed Gateway.

1. Enter the IP address for the object, and select the **Firewall** option.

There is no need to define the exact topology of each externally managed Gi Gateway/Member. In the case of a Gi cluster, the IP address used should be the unique IP of the Cluster Member, and not the IP address of the cluster itself.

2. Add the Gi members into the Overbilling group.

## On the Carrier Security Gateways

1. Set SAM to use SSL on Carrier Security Gateways by updating the file

`$CPDIR/conf/sic_policy.conf`.

- a. Use a text editor to open the file, and search for the line `[Outbound rules]`.
- b. Insert a new line with this format:

```
ANY ; "DN" ; ANY ; sam ; ssl
```

"DN" is the unique name of each Gi Gateway/Member, as it appears in the Gi SmartConsole, on the main page of the Gi object.

For example, if the name of the Gi management is `gi-mgmt`, and the name of one of the Gi Gateways/Members is `gi-mod1`, the DN would be something like `"CN=gi-mod1,O=gi-mgmt..7au2cw"`.

And so, the line in `sic_policy.conf` would look like:

```
ANY ; "CN=gi-mod1,O=gi-mgmt..7au2cw" ; ANY ; sam ; ssl
```

**Note** - The double quotes in the line are mandatory. Be sure to use double quotes ("), and not single quotes (') when writing the line in `sic_policy.conf`.

For every additional Gi gateway/Member you wish to use, add additional lines below the lines you have just added. Be sure to use the correct DN for each new Gi gateway.

2. Establish a trust relationship between Security Gateways by running this command on each Carrier Security Gateway/Member:

```
fw putkey -ssl -p [<secret>] [<IP address of Gi
gateway/Member>]
```

[<secret>] is any string that will be used in the first authentication between the Carrier Security and the Gi/SGi Gateways. The string used here must match the string used in the "fw putkey" command which you run on the Gi/SGi Gateway/Member.

For additional Gi/SGi Gateways/Members, run the "fw putkey" command again with the IP address of that member.

Make sure that in all cases you use the unique IP address of each cluster member, and not the IP address of the cluster itself.

3. Run `cpstop` and `cpstart` on all Carrier Security Gateways/Members on which you have edited `sic_policy.conf` for the changes to take effect.

### On the Gi/SGi Management

1. Define a node object using the IP address of the Carrier Security cluster. Note that you need to use the Carrier Security cluster IP address associated with the interface facing the Gi Gateway/Cluster.
2. Define a rule allowing FW1\_sam traffic from the Carrier Security cluster IP address to the Gi Gateways/Members.

| Source                   | Destination      | Service | Action |
|--------------------------|------------------|---------|--------|
| Carrier Security Gateway | Security Gateway | FW1_sam | Accept |

3. Install the policy on the Carrier Security members.

### On the Gi/SGi Gateways:

1. Set SAM to use SSL on Security Gateways by updating the file `$CPDIR/conf/sic_policy.conf`.
  - a. Use a text editor to open the file, and search for the line `[Inbound rules]`.
  - b. Insert a new line with this format:

```
ANY ; "DN" ; ANY ; sam ; ssl
```

"DN" is the unique name of each Carrier Security Gateway/Member, as it appears in the Carrier Security SmartConsole, on the main page of the Carrier Security Security Gateway / Cluster object.

For example, if the name of the Management Server is "gx-mgmt", and the name of one of the GX Gateways/Members is "gx-mod1", the DN would be something like "CN=gx-mod1,O=gx-mgmt..7au2cw".

And so, the line in `sic_policy.conf` would look like:

```
ANY ; "CN=gx-mod1,O=gx-mgmt..7au2cw" ; ANY ; sam ; ssl
```

Note - The double quotes in the line are mandatory. Be sure to use double quotes ("), and not single quotes (') when writing the line in `sic_policy.conf`.

For every additional Carrier Security Gateway/Member, add additional lines below the previous lines you've added. Be sure to use the correct DN for each new Carrier Security Gateway.

2. Establish a trust relationship between Security Gateways by running this command on each Gi Gateway/Member:

```
fw putkey -ssl -p [<secret>] [<IP address of Carrier Security Gateway/Member>]
```

Where [<secret>] is any string that will be used in the first authentication between the Carrier Security and the Gi Gateways. The string used here must match the string used in the "fw putkey" command which you run on the Carrier Security Gateway/Member.

For additional Carrier Security Gateways/Members, run the "fw putkey" command again with the IP address of that member.

Make sure that in all cases you use the unique IP address of each member, and not the IP address of the shared cluster.

3. Run `cpstop` and `cpstart` on all Gi/SGi Gateways/Members on which you have edited `sic_policy.conf` for the changes to take effect.

## Testing Overbilling Protection

Enabling Overbilling protection for a clustered environment is complex, and is prone to mistakes. It is therefore highly recommended to perform several tests to make sure the configuration was done correctly.

1. Delete a GTP tunnel. On the CS management, examine the logs and verify that the GTP tunnel deletion was done through a specific Carrier Security member.
2. On the Gi/SGi management, verify that there is a log from each Gi/SGi Gateway/Member reporting that a SAM rule has been added.
3. Delete a GTP tunnel through the other CS members. In most cases, you will need to perform a failover in the CS cluster.

4. Again verify that there is a log from each Gi Gateway/Member reporting that a SAM rule has been added.

## Disabling Broadcast Inspection

When you create a tunnel GTP tunnel, there is an exchange of **Create PDP context (GTPv1)** or **Create Session (GTPv2)** messages. During this exchange, the mobile subscriber PDP address (IP address used to access the PDN) is set.

By default, GTP inspection makes sure that a PDP address is not a broadcast address. Because some customers use an IP network from one class, but work with it as if it was from a different class. This can cause the use of broadcast IP addresses from the original class, which are dropped by broadcast inspection. You must disable broadcast inspection to resolve this issue.

To disable PDP address broadcast Inspection for the current session (until reboot), run:

```
fw ctl set int gtp_check_eu_broadcast_address 0
```

To permanently disable PDP address broadcast inspection, add this line to the `$FWDIR/boot/modules/fwkernel.conf` file: `gtp_check_eu_broadcast_address=0`

## Enforcing a More Granular GTP Security Policy

Through the use of GTP service filters and APN objects, you can create security rules that allow GTP traffic only within the parameters that you specify. GTP-aware filters provide the ability to limit the creation of PDP contexts to traffic that matches specific parameters that you define.

## Customizing GTP Services

Carrier Security allows you to build your own GTP service filters, which can provide a more granular GTP security policy. When you create a new GTP service, you can set exactly which prefixes or other identifiers to allow onto your network. While you can modify the pre-defined GTP services `gtp_v0_default` and `gtp_v1_default`, it is recommended that you create a new service. As such, the following GTP customization example begins with the steps required to create a new service.

To define a new GTP service:

1. In **SmartConsole** open **Objects > Object Explorer**.
2. Click **New > Service > GTP**
3. On the **General** page:
  - a. Give the new GTP service filter a name.

**Note** - The port numbers of the GTP services cannot be modified.

- b. For **GTP Message Filtering**, select an **Interface Profile**.

GTP Message Filtering Specifies GTP interface profile which defines associated control messages that match this service. You can create a custom profile or manually customize a service to filter specific message types.

- c. Select **Actions**.

- **Allow usage of static IP addresses** for mobile subscribers with pre-assigned IP addresses. While IP addresses are usually allocated by the GGSN, some users may have static, pre-assigned IP addresses. The default is to allow such paths. When this option is set, PDP context activation will be enabled in static mode as well.
- **Apply Access Policy on user traffic** causes all mobile user traffic encapsulated in G-PDUs to be inspected by FireWall and IPS stateful inspection. *This prevents GTP-U acceleration and requires consideration when allocating CPU cores for CoreXL instances and SND.*
- **Add IMSI field to logs generated by user traffic** inserts the value in the IMSI field for any log generated by mobile user data, linking the log to the mobile user.

4. Click **Advanced**.

For parameter, specify a value or select **Any**.

- **IMSI Prefix** specifies a subscriber identity prefix. The subscriber identity prefix is usually of the form Country and Operator, for example, 23477 (where 234 is the MCC and 77 is the MNC).
- **Access Point Name** specifies an APN object. An example of an APN is internet.mnc55.mcc243.gprs, or example.com. For APN configuration information, see ["Creating an APN Object" on the next page](#)
- **Selection Mode** specifies a selection mode indicating the origin of the APN that appears in the PDP context request.
- **MS-ISDN** specifies an MS-ISDN prefix (for example, 447788).
- **LDAP group** specifies an LDAP group, sorted by two main attributes.

**According to MS-ISDN or IMSI** identifies whether a user belongs to a specific LDAP group by IMSI or MS-ISDN.

5. Click **Radio Access Technology**.

Specify which radio types a request has to match. You can customize the service to perform these actions on matching GTP traffic.

6. Click **Additional Services**.

On the **General** page, if you selected **V2** as the version, select one or more of these additional service types:

- **Trace Management**
- **CS Fallback and SRVCC**
- **Restoration and Recovery**

Add a rule in the rule base using this service and make sure the rule is above all other GTP-based rules.

## Creating an APN Object

It is possible to control which end user IP addresses can access your network by enforcing APN domains. This is useful in environments where some or all of the local APNs have a pre-defined unique domain of end user IP addresses. To create an APN object:

1. In **SmartConsole** open **Objects > Object Explorer**.
2. Click **New > Network Object > More > Access Point Name**.

The **APN Properties** window shows.

The screenshot shows the 'Access Point Name' configuration window. The title bar says 'Access Point Name'. Inside, there's a header with a mobile phone icon and the text 'apn\_Jamaica' and 'Enter Object Comment'. Below this, the 'General' tab is active, showing a text field for 'APN string' with the value 'internet.mnc55.mcc.gprs'. The 'Intra tunnel policy enforcement' section is expanded, showing three checked options: 'Enforce end user domain' (with a dropdown set to 'Jamaica'), 'Block MS to MS traffic between this and other APN end user domains', and 'Block MS to MS traffic within this end user domain'. At the bottom, there's an 'Add Tag' button and 'OK' and 'Cancel' buttons.

3. Name the APN.
4. Specify an **APN string** that identifies the APN, for example `internet.mnc55.mcc243.gprs`, or `example.com`.

- Define a GTP service that is restricted to the APN you defined in the previous step. Define the new GTP service in the **GTP Service Properties** window and set its properties in the **Advanced GTP Service Properties** window. If the end user's IP address does not match the APN string, then PDP context create response or request messages are dropped with the error message `IP is not in the APN domain`.

To match more than one string to an APN rule:

- Create an APN.
- Include a wildcard in its name, such as `*.example.gprs`. An APN with this name will match strings with names like `123.example.gprs` and `abc.example.gprs`. Supported wildcards are:

| Wildcard | Explanation                  |
|----------|------------------------------|
| *        | any number of any characters |
| ?        | any 1 character              |

## Using APNs in a Security Policy

Suppose two APNs are defined in this way:

### APN End User Domain example

| Name | IP address range     | Block MS to MS traffic between this and other APN End User Domains | Block MS to MS traffic within this APN End User Domain |
|------|----------------------|--------------------------------------------------------------------|--------------------------------------------------------|
| APN1 | 10.1.1.0 - 10.1.1.24 | checked                                                            | checked                                                |
| APN2 | 20.1.1.0 - 20.1.1.24 | not checked                                                        | not checked                                            |

G-PDUs encapsulated in PDP-contexts using APN\_Jamaica with server IPs from the range 10.1.1.0/24 or 20.1.1.0/24 will be dropped.

No restriction will be placed on G-PDUs belonging to APN\_Spain. Specifically, a packet sent from a server to an MS with source IP 10.1.1.4 and destination IP 20.1.1.7 is allowed.

For more information on configuring APNs, see: ["GTP Intra Tunnel Inspection and Enforcement" on the next page](#).

## Blocking MS to Gn/S5 Network Traffic

To deny MS to Gn/S5 traffic:

1. Define an APN object for the Gn/S5 network (the APN string value in this case is not important).
2. Enable the property **Enforce End User Domain** and select the domain with the range of IP addresses of the Gn/S5 network you want to protect.
3. Enable the property **Block MS to MS traffic between this and other APN End User domains** on this and all other APNs defined.

## GTP Intra Tunnel Inspection and Enforcement

### Defining Access Using APN Objects

To define Intra Tunnel enforcement by end user domain:

1. Create an APN object as detailed in "Creating an APN Object", or edit an existing APN object.
2. Enable Enforce End User Domain to block user traffic that is outside a range of defined IP addresses.
3. Choose the relevant End User Domain from the drop down list.
4. Enable Block MS to MS traffic between this and other APN End User domains to drop and log intra-tunnel traffic between two MSs with IP addresses `other than the ones` specified in this APN End User Domain drop-down list.
5. Enable Block MS to MS traffic within this APN domain to drop and log intra-tunnel traffic between two MSs with IP addresses `that match` the addresses specified in this APN End User Domain drop-down list.

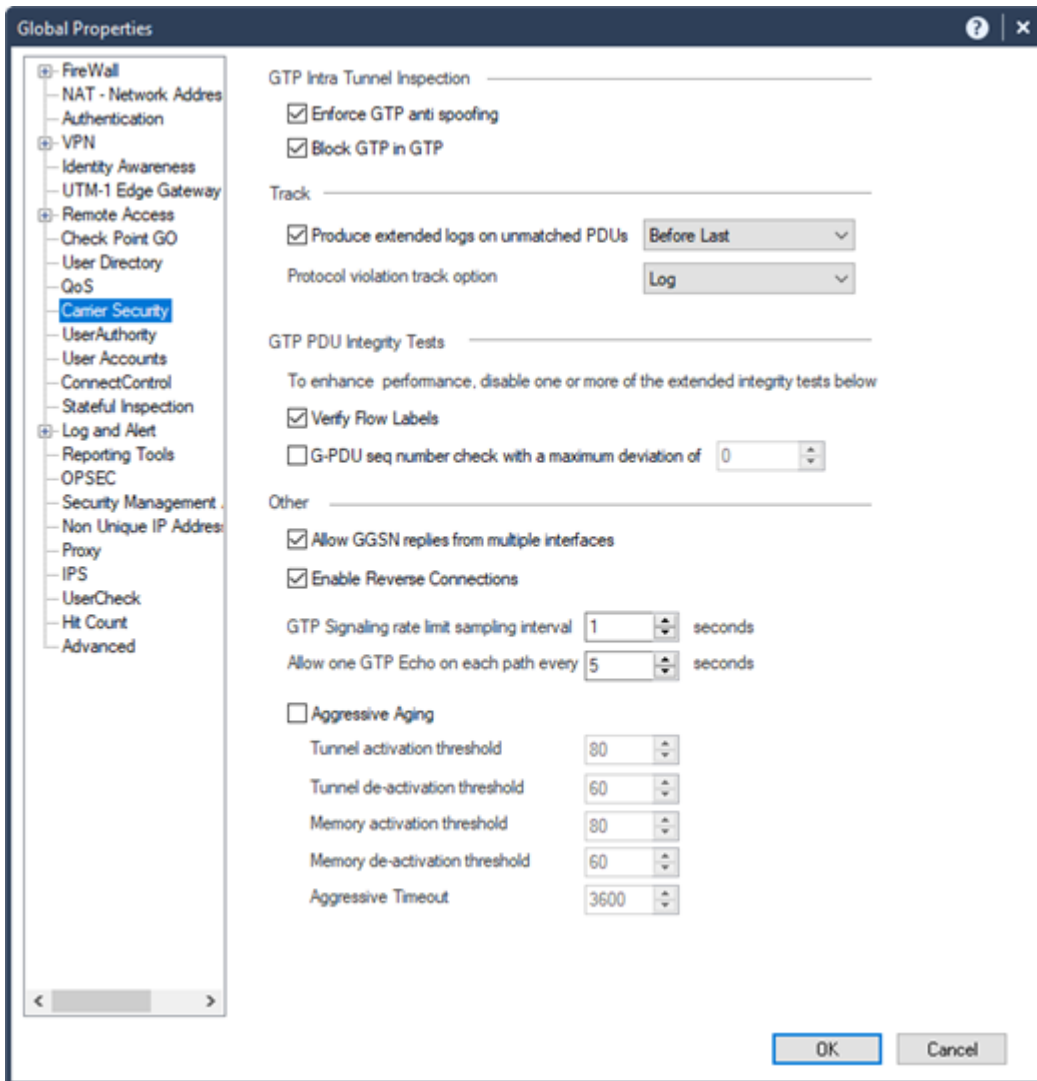
For conceptual information, see: ["APN Domain End User Address Enforcement" on page 33](#).

### Enforce GTP Anti-Spoofing

GTP Anti-Spoofing drops G-PDUs that do not use the end user IP address agreed upon in the PDP context activation process. The setting Enforce GTP Anti-Spoofing is located in the Carrier Security tab of the Global Properties window. Its default setting is enabled.

### Block GTP in GTP

**Block GTP in GTP** drops GTP packets encapsulated inside GTP tunnels. The setting is located in the **Carrier Security** tab of the **Global Properties** window, and enabled by default.



## GTP PDU Integrity Tests

These GTP PDU Integrity checks are available on the **Carrier Security** tab of the **Global Properties** window:

- **Verify Flow Labels** checks that each packet's flow label matches the flow labels defined by GTP signaling. This option is relevant for GTP version 0 only. The default setting is unchecked.
- **G-PDU seq number check with a maximum deviation of** a value set here. Sequence checking is enforced, but an out-of-sequence G-PDU is accepted if the difference between its sequence number and the expected sequence number is less than or equal to the maximum deviation. The default setting is unchecked.

The following related parameters take effect only if **G-PDU sequence number check with a maximum deviation of** is enabled, and can be configured using the [Database Tool \(GuiDBEdit Tool\)](#):

- `gtp_sequence_deviation_drop` - Drop all out-of-sequence packets. The default setting is `FALSE`.
- `gtp_sequence_deviation_alert` - Generate a log when an out-of-sequence packet is encountered. The default setting is `TRUE`.

**Note** - GTP PDU Integrity Tests are not supported in accelerated mode.

## Secure Connectivity Features

The following features are concerned with connectivity:

- **Allow GGSN Replies From Multiple Interfaces** - for GTP signaling replies from an IP address different from the IP address to which the requests are sent. The default setting is checked.

When this option is enabled, be sure to protect against Session Hijacking through the use of Handover Groups. For information on setting up Handover Groups, see ["Creating a GSN Handover Group" on page 36](#)

- **Enable Reverse Connections (for Gateways below version R80)** - accepts PDUs from the GGSN to the SGSN on a previously established PDP context even if these PDUs are sent over ports that do not match the ports of the established PDP context.

This is available for the following PDUs:

- G-PDUs.
- Delete PDP context/sessions requests.
- GTP version 1 update PDP context requests.
- GTP version 2 update/delete bearer.
- GTP error indication message.

The default setting is `enabled`.

These features are located in the **Other** section of the Carrier Security tab in **Global Properties**.

To enable such connections on Security Gateways that run R80.20 and higher, manually configure a reverse rule such as this:

| Source      | Destination | VPN | Services & Applications                            | Action | Track | Install On     | Comment                                    |
|-------------|-------------|-----|----------------------------------------------------|--------|-------|----------------|--------------------------------------------|
| PartnerA_HG | GG_Home_HG  | Any | gtp_v0_default<br>gtp_v1_default<br>gtp_v2_default | Accept | Log   | Policy Targets | Roaming for my users on partner's networks |
| SG_Home_HG  | PartnerA_HG | Any | gtp_v0_default<br>gtp_v1_default<br>gtp_v2_default | Accept | Log   | Policy Targets | Roaming for partner's users on my network  |

| Source                        | Destination                   | VPN | Services & Applications                            | Action | Track | Install On     | Comment                                          |
|-------------------------------|-------------------------------|-----|----------------------------------------------------|--------|-------|----------------|--------------------------------------------------|
| GG_Home_HG<br><br>PartnerA_HG | PartnerA_HG<br><br>SG_Home_HG | Any | gtp_v0_reverse<br>gtp_v1_reverse<br>gtp_v2_reverse | Accept | Log   | Policy Targets | Permitting reverse connections for both purposes |

## Limiting Signaling Rates

- **Allow one GTP Echo on each path every x seconds** sets the interval at which GTP Echo requests are allowed per path. Echo requests exceeding this rate will be dropped and logged. You can disable the signaling rate limit, and thereby accept all Echo requests, by entering 0. The default setting is 1.
- **GTP Signaling rate limit sampling interval** sets the interval for signal packet rate sampling. The default setting is 1 second.
- **Enforce GTP Signal packet rate limit to host** sets the number of PDUs allowed per second. PDU traffic exceeding this rate will be dropped and logged. This feature protects local GSNs from Denial of Service attacks by restricting the signaling rate that can be sent to a local GSN. Configure this rate on the **Carrier Security** page of each GSN network object. If checked, GTP signaling PDUs destined to this GSN above the specified rate are blocked and dropped. The default rate is 2048.

Consider the following example: The rate limit sampling interval is set to the default rate of 1 second and the network object has enforced a GTP signal packet rate limit of the default of 2048 PDU per second. Then sampling will occur once per second and will allow 2048 signaling PDUs between two successive samplings.

The following related parameters can be set using the [Database Tool \(GuiDBEdit Tool\)](#):

- `gtp_rate_limit_drop` drops packets that exceed the configured rate. The default value is `TRUE`.
- `gtp_rate_limit_alert` issues an alert when packets exceed the configured rate. The default value is `TRUE`.
- **Enforce GTP Signal packet rate limit from GSN handover group** sets the number of PDUs allowed per second *from* a GSN handover group object. And protects the network core from flooding.

## Using FW SAM to Close PDP Contexts/sessions

For more information about the "fw sam" command on Security Gateway, see the [R82 CLI Reference Guide](#) > Chapter *Security Gateway Commands* > Section *fw* > Section *fw sam*.

### Carrier Security Content Filter Arguments (<key=val>) for the "fw sam" Command

**Note** - Key names and their values are case sensitive.

| Key        | Value                               | Meaning                                                                                                                                                                                                                           |
|------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| service    | gtp                                 | Service of 'gtp' indicates that the request applies only to connections that go through the gtp tunnel between the SGSN and the GGSN machines.<br>All Carrier Security requests must include this argument.                       |
| imsi       | String of numbers                   | International Mobile Subscriber Identity, a unique number identifying a particular mobile subscriber which comprises the MCC (Mobile Country Code), MNC (Mobile Network Code) and MSIN (Mobile Subscriber Identification Number). |
| msisdn     | String of numbers                   | Mobile Subscriber phone number.                                                                                                                                                                                                   |
| apn        | String of up to 115 characters      | Access Point Name.                                                                                                                                                                                                                |
| tunl_dst   | Dotted decimal IP address           | Destination IP address of the tunneled connection                                                                                                                                                                                 |
| tunl_dport | Short integer represented as string | Destination port of the tunneled connection                                                                                                                                                                                       |
| tunl_proto | Short integer represented as string | Destination protocol of the tunneled connection                                                                                                                                                                                   |

The next table lists the different types of Carrier Security requests, each represented with a certain combination of key=value pairs.

### Carrier Security request types

**Note** - It is not possible to monitor the Carrier Security requests for the "-M" option. Key names and their values are case sensitive.

| Request Type                | Includes                         | Notes |
|-----------------------------|----------------------------------|-------|
| Destination Network Request | <b>APN</b>                       |       |
| User Identification         | <b>IMSI</b> and/or <b>MSISDN</b> |       |

| Request Type        | Includes                                                                                                                                                                                                                      | Notes                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User to Destination | 1) <b>IMSI</b> and/or <b>MSISDN</b><br>2) One or more of the following destination arguments:<br><b>APN</b><br>All of these tunneled connection arguments together: <b>tunl_dst</b> , <b>tunl_dport</b> and <b>tunl_proto</b> | The three tunneled connection arguments of <b>tunl_dst</b> , <b>tunl_dport</b> , and <b>tunl_proto</b> , must come together.<br>The Carrier Security Gateway does not close open tunnels.<br>Therefore, a request that includes <b>tunl_dst</b> , <b>tunl_dport</b> , and <b>tunl_proto</b> may not be used with "-J" and "-I" options. |

These examples demonstrate the use of the generic criteria for sending a Carrier Security request:

| Scenario                                               | Example FW SAM command                                                                                                                                                                                                                                                           |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| APN only                                               | <code>fw sam -f monica-gx -I generic service=gtp apn=test.com</code>                                                                                                                                                                                                             |
| IMSI (and/or MSISDN) only                              | <code>fw sam -f monica-gx -I generic service=gtp imsi=055123456 (for MSISDN: msisdn=380561234567)</code>                                                                                                                                                                         |
| IMSI (and/or MSISDN) and APN:                          | <code>fw sam -f monica-gx -I generic service=gtp apn=test.com imsi=055123456</code>                                                                                                                                                                                              |
| IMSI (and/or MSISDN) and 3-tunnel connection arguments | <code>fw sam -f monica-gx -i generic service=gtp imsi=055123456 tunl_dst=10.100.100.1 tunl_dport=80 tunl_proto=6</code><br>This call inhibits connections on Carrier Security object monica-gx from a user with IMSI number 05123456 connecting over HTTP to server 10.100.100.1 |

### Carrier Security SAM request to deal with unusable GTP tunnels

You can use a special SAM request in the event that one or more GTP tunnels are suspected to be hanging. This can occur due to connectivity failure between SGSNs and GGSNs when waiting for these tunnels to expire is not an option.

**Note** - Carrier Security deletes tunnels automatically after the default time out value of 90,000 seconds (25 hours).

When you try to delete such tunnels using regular SAM requests, the Carrier Security module sends Delete PDP Context request messages to the SGSNs and GGSNs related to the specific tunnel, which may not be answered due to connectivity failure. Thus, these tunnels will not be deleted.

To delete these tunnels without sending the Delete PDP Context requests, a global variable has to be set before initiating the special SAM request, and afterwards - unset.

To delete unusable tunnels, run these commands in the Expert mode on the Carrier Security Command Line:

1. `fw ctl set int allow_sam_delete_gtp_tunnels 1`
2. `fw sam -f monica-gx -t 1 -J generic service=gtp imsi=055123456`  
(or any other combination as described in the table above)
3. `fw ctl set int allow_sam_delete_gtp_tunnels 0`

### Adding Support for New GTP Messages and Information Elements

New TLV Information Elements and/or GTP Message Types can be added to the list of messages recognized by Carrier Security, thereby allowing them to pass through the firewall.

To define the new Elements or Types, add the relevant line(s) to the end of file `$FWDIR/lib/gtp.def` on the Management Server:

```
gtp_additional_types = {new GTPv0/1 Message Types};

gtpv2_ignore_messages = { new GTPv2 message types};

gtp_additional_elements = {new Information Elements};

gtpv2_ignore_elements = {new gtpv2 information elements};
```

Each line should list the ID (number) of the additional Message Types and/or Information Elements, respectively.

For example: if you define the following:

```
gtp_additional_types = {71, 72, 73};

gtp_additional_elements = {239, 240, 241};
```

the Message Types 71, 72 and 73 and Information Elements 239, 240 and 241 will be allowed to pass through the system when GTP version is 0/1, for GTPv2 use the other tables.

As long as their GTP headers are valid, the new Message Types will pass irrespective of their content. The new Information Elements defined may be included in any Message Type, and can appear in any location in the sequence of Information Elements in the message. You may add just new Message Types, or just new Informational Elements, or both for each of the versions.

## Adjusting Settings with Database Tool (GuiDBEdit Tool)

There are a number of settings within Carrier Security that do not have a Graphical User Interface. The value of these properties may be changed using the [Database Tool \(GuiDBEdit Tool\)](#). Global settings are detailed in the next table:

### Global Settings

| Property                     | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Default |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| gtp_sequence_deviation_drop  | If T-PDU sequence number check with a maximum deviation of is enabled, drop out-of-sequence packets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | FALSE   |
| gtp_sequence_deviation_alert | If T-PDU sequence number check with a maximum deviation of is checked, a log will be generated when an out-of-sequence packet is encountered.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | TRUE    |
| gtp_allow_recreate_pdpc      | When a Create PDP Context arrives at the Carrier Security Gateway and the tunnel already exists, the question whether this new Create should be allowed depends on whether the Carrier Security Gateway is configured to be strict or open with regard to this scenario.<br>For GTP Version 1/2, a tunnel is composed of four TEIDs. If any one of the four TEIDs of a new create attempt is already in use (for the same GSNs pair), this will be considered a recreate. If gtp_allow_recreate_pdpc is set to open, the recreate is allowed. The Create Log generated for the new tunnel will include a remark in the info field stating "reusing TEID". | OPEN    |
| gtp_rate_limit_drop          | A packet exceeding the allowed rate is dropped by default. To accept such packets, change the property's value to FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | TRUE    |
| gtp_rate_limit_alert         | If a packet exceeds the allowed rate, a log is issued. To cancel such logs, change the property's value to FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | TRUE    |
| gtp_chk_hdr_len              | If TRUE, Carrier Security verifies the length written in the GTP header.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | TRUE    |
| gtp_delete_upon_error        | If TRUE, an error on a tunnel causes the tunnel to be deleted from the Carrier Security tables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | FALSE   |

| Property                      | Meaning                                                                                                                                                                                                                                                                                                                  | Default |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| gtp_echo_requires_path_in_use | If TRUE, Carrier Security verifies that at least one tunnel between the SGSN and GGSN participating in the echo is established.                                                                                                                                                                                          | FALSE   |
| gtp_loggrace                  | Carrier Security eliminates similar logs indicating error each gtp_loggrace seconds.                                                                                                                                                                                                                                     | 10      |
| gtp_max_req_retransmit        | Only gtp_max_req_retransmit retransmissions of the same request are allowed.                                                                                                                                                                                                                                             | 5       |
| gtp_monitor_mode              | If TRUE, Carrier Security will not drop any GTP traffic even if it was evaluated as malicious, illegal, etc. The CS logging system will however log the intended drop as it would in regular operation mode. This enables the operator to realize the impact of CS on the system without actually enforcing that impact. | FALSE   |
| gtp_log_additional_fields     | If TRUE, additional Information Elements are added to the logs of GTP traffic.                                                                                                                                                                                                                                           | FALSE   |

### Settings per Gateway

| Property                  | Meaning                                                                                                                                                                                                                             | Default    |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| gtp_pending_hashsize      | Sets the hash size of the gtp_pending kernel table, which is used to store pending GTP signaling requests. This value must be a power of 2.                                                                                         | 65536      |
| gtp_pending_limit         | Sets the maximum number of entries stored in the gtp_pending kernel table.                                                                                                                                                          | 25000      |
| gtp_pending_timeout       | Sets the timeout of entries in the gtp_pending kernel table.                                                                                                                                                                        | 40 seconds |
| gtp_sam_close_upon_delete | A Boolean parameter used to enable sending a delete PDP context request message to GSNs when a tunnel is deleted using the SAM API or when PDP contexts expire. Enabled automatically when the CS Overbilling protection is in use. | FALSE      |

| Property                     | Meaning                                                                                                                             | Default          |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------|
| gtp_<br>tunnels_<br>hashsize | Sets the hash size of the gtp_tunnels kernel table, which is used for storing active PDP contexts. This value must be a power of 2. | 65536            |
| gtp_<br>tunnels_<br>limit    | Sets the maximum number of entries stored in the gtp_tunnels kernel table.                                                          | 50000            |
| gtp_<br>tunnels_<br>timeout  | Sets the timeout of entries in the gtp_tunnels kernel table.                                                                        | 90000<br>seconds |

# Monitoring GPRS Network Security

To effectively manage your network and make informed decisions, you need to gather information on the network's traffic patterns. If you experience connectivity or security related problems, you need to be able to identify changes in the traffic. Carrier Security provides a set of tools that address the special needs of cellular networks.

## GTP Tracking Logs and Alerts

Carrier Security records cellular-specific information of GTP signaling activity, including APN, IMSI, Selection Mode, GSN addresses and more. The information recorded in these logs can help you determine why certain GTP traffic may be dropped or rejected, and to decide if whether to adjust the Security Policy to accept that traffic.

The Carrier Security GTP Inspection Gateway generates a wide range of detailed security alerts in the event of protocol and Security Policy violations, including PDU details, network information and protocol violation type. Carrier Security also provides GTP-specific alerts on malformed packets and malicious activity. For information on setting the alert type, see ["Configuring Monitoring" on the next page](#)

**Note** - As in all security rules, you must set the tracking option of the rule to Log if you want to record GTP activity.

## Recording GTP Data from Unmatched PDUs

Carrier Security can record GTP traffic that is not matched by a GTP rule in the rule base. Normally, traffic that is not matched is logged in the general log as a simple `Drop`. Carrier Security provides a tool for capturing this data with the special GTP-related fields that can help to discover the cause of these drops. To configure this feature, see: ["Configuring Monitoring" on the next page](#)

## GTP Accounting

By setting a GTP user traffic rule to `Log`, Carrier Security generates a log entry for every terminated PDP context that matches on the rule. The log records the total number of user packets (`n_pdu`) and bytes (`n_byte`) transferred in the user plane during the PDP context. Carrier Security issues logs for the following events:

- PDP context/session delete
- Tunnel expiration
- Tunnel recreation
- Active Gateway goes down (when in High Availability mode)

## Excessive Logs Protection

Due to the small packet nature of cellular communications, Carrier Security records a vast amount of data each day, far more than a typical Check Point firewall. This data collection is essential to accurately diagnose network status, and to troubleshoot network errors.

This intensive logging activity could make some systems more vulnerable to Denial of Service (DoS) attacks. Carrier Security protects against this type of attack by setting a `similar logging` threshold, above which it does not generate similar logs. This feature is configurable. See `gtp_loggrace` in ["Adjusting Settings with Database Tool \(GuiDBEdit Tool\)" on page 56](#).

The default is every 10 seconds.

## Monitor-Only Mode

Monitor-Only Mode tracks certain unauthorized traffic without blocking it. While in this mode, the firewall continues to inspect GTP traffic, but does not enforce any of the GTP related protections. It does continue to enforce GTP-related security rules, log GTP-related activity, and issue GTP error logs and alerts. Monitor-Only Mode enables operators to preview the results of changes to global properties and settings concerning GTP inspection. This mode is helpful in preventing unanticipated behavior when phasing in Carrier Security for the first time, and whenever changes are made to the global properties.

After a careful review of the logs and ensuring that the changes do not impede legitimate cellular traffic, the cellular operator can turn off Monitor-Only Mode, and the firewall can commence blocking malicious GTP traffic.

Carrier Security follows the GTP tunnels and keeps their `state` as it would in regular operation mode. Therefore you can smoothly switch Monitor-Only Mode on and off - all tunnel information continues to exist in both modes, and no tunnels are lost in transition.

For configuration information, see `gtp_monitor_mode` in: ["Adjusting Settings with Database Tool \(GuiDBEdit Tool\)" on page 56](#)

## Configuring Monitoring

- **Produce extended log on unmatched PDUs** logs GTP packets not matched by previous rules with Carrier Security's extended GTP-related log fields. These logs appear brown and their Action attribute is empty. The default setting is `checked`.
- **Protocol violation track** option allows you to set the appropriate track or alert option to be used when a protocol violation (malformed packet) is detected. The default setting is `Log`.

## Monitoring GSN Handover Group Limits

You can enable these options in **Global Properties > Carrier Security > Track**.

Use this command to see tunnel use for handover groups. Run the command in the Expert mode on the Security Gateway.

### Syntax

```
fw gtp ho_groups [-s { name | tunnels | limit | util } [-r]] [-m <lines>]
```

```
fw gtp ho_groups -g <name>
```

```
fw gtp ho_groups -l
```

```
fw gtp ho_groups -h
```

| Parameter  | Description                                                          |
|------------|----------------------------------------------------------------------|
| -s         | Sort the output by tunnel name, assigned limit or tunnel utilization |
| -r         | Sort the output in in reverse alphabetical order                     |
| -m <lines> | Show, at most, the specified number of lines                         |
| -l         | Show only the handover group names (no data)                         |
| -g <name>  | Show only the specified handover group                               |

### Example:

```
# fw gtp ho_groups
Name Open tunnels Limit %Utilization
-----
--
Operator-6-GSNs 25000 100000 25
Operator-9-GSNs 33148 50000 66
Operator-3-GSNs 380 no limit n/a
Operator-8-GSNs 15897 200000 7
Operator-5-GSNs 84125 180000 46
Operator-4-GSNs 0 50000 0
Operator-1-GSNs 45000 45000 100
Operator-7-GSNs 69716 70000 99
Operator-2-GSNs 394326 500000 78
```

# SNMP Extensions for GTP Statistics

Carrier Security can be configured to send SNMP polling data and traps. Various cellular-specific statistics, such as the number of PDP contexts created and deleted, can be polled by SNMP monitoring stations. Alerts and logs can also be set via GTP-aware security rules to send SNMP traps to a monitoring station when events that require attention occur.

The Check Point MIB is a data file that contains the collection of Check Point's SNMP-manageable network devices. It contains SNMP extensions for Carrier Security.

For more information about SNMP and MIB, see the [R82 Gaia Administration Guide](#).

The Check Point MIB file can be found on the Security Gateway in the `$CPDIR/lib/snmp/` directory.

## Understanding Check Point OIDs:

- Prefix OID for Check Point root is: **1.3.6.1.4.1.2620**. (Check Point is 2620)
- Prefix OID for **GX** root is: **1.3.6.1.4.1.2620.1.20**. (Products is 1, GX is 20)

## GX SNMP Counters Tree

### GX (20)

gxProdVerMajor (5.2)

gxProdVerMinor (5.3)

gxBuild (5.4)

### gxInfo(1)

gxProdName (1)

gxProdVersion (2)

### gxCreateInfo(5)

gxCreateSinceInstall (1)

gxActContxt (2)

gxDropPlicyCreate (3)

gxDropMalformedReqCreate (4)

gxDropMalformedRespCreate (5)

gxExpiredCreate (6)

gxBadCauseCreate (7)

gxSecondaryNsapiEntries (8)

### gxDeleteInfo (6)

gxDeleteSinceInstall (1)  
 gxDropOutOfContxtDelete (2)  
 gxDropMalformedReqDelete (3)  
 gxDropMalformedRespDelete (4)  
 gxExpiredDelete (5)  
 gxBadCauseDelete (6)

**gxUpdateInfo (7)**  
 gxUpdateSinceInstall (1)  
 gxDropOutOfContxtUpdate (2)  
 gxDropMalformedReqUpdate (3)  
 gxDropMalformedRespUpdate (4)  
 gxExpiredUpdate (5)  
 gxBadCauseUpdate (6)

**gxPathMngInfo (8)**  
 gxEchoSinceInstall (1)  
 gxVnspSinceInstall (2)  
 gxDropPolicyEcho (3)  
 gxDropMalformedReqEcho (4)  
 gxDropMalformedRespEcho (5)  
 gxExpiredEcho (6)  
 gxDropVnsp (7)  
 gxGtpPathEntries (8)

**gxGpduInfo (9)**  
 gxGpdu1MinAvgRate (1)  
 gxDropOutOfContxtGpdu (2)  
 gxDropAnti-spoofingGpdu (3)  
 gxDropMs-MsGpdu (4)  
 gxDropBadSeqGpdu (5)  
 gxDropBadGpdu (6)  
 gxGpduExpiredTunnel (7)

## Example

gxActContxt SNMP counter OID is: (GX Active Contexts - gtp\_tunnels counter)



## Testing SNMP Functionality

To test the Carrier Security's SNMP internal functionality, use this command on the module side:

```
gxstattest <oid>
```

# Log Messages

Check Point products provide you with the ability to collect comprehensive information on your network activity in the form of logs. You can audit these logs at any given time, analyze your traffic patterns and troubleshoot networking and security issues. Familiarizing yourself with the logs can help you understand and learn the status of your network, as well as resolve problems you are experiencing with the system. Reviewing traffic logs is a very important aspect of security management, and should get careful attention.

## Carrier Security Log Messages

This section contains a list of Carrier Security log messages and resolutions, listed alphabetically. You may encounter self-explanatory log messages that are not included here.

| Log Message                                | Meaning                                                                                                                        | Resolution                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duplicate sequence number                  | This G-PDU carries a duplicated sequence number.                                                                               | Enforcement of G-PDU sequence numbers is determined in the Carrier Security page of the Global Properties window.<br>Also, it is possible to change the drop and alert behavior of the rate limiting feature by editing the <code>gtp_sequence_deviation_drop</code> and <code>gtp_sequence_deviation_alert</code> properties with the GUI Dedit tool. |
| Echo Request not within time limit         | An echo request was received too close to a previous echo request. This echo request will be dropped.                          | This will happen only if you set the value of the <code>gtp_echo_frequency</code> property to the number of seconds required between Echo Requests. You can use this parameter to protect against Echo Request Flooding.                                                                                                                               |
| Echo Request on a path which is not in use | An echo request was received on a path (SGSN-GGSN pair) that currently has no active PDP Context. The request will be dropped. | This happens if you set the value of the <code>gtp_echo_requires_path_in_use</code> property. By default such Echo Requests are not dropped.                                                                                                                                                                                                           |

| Log Message                                             | Meaning                                                                                                                                                                         | Resolution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GTP quota threshold alert: too many packets             | This packet (PDU) exceeded the Signaling Rate Limit defined for the indicated destination host                                                                                  | This could be the result of a Signaling flood attack. If this happens during normal operation it might be advisable to increase Enforce GTP Signal packet rate limit for this GSN entity in the Carrier Security page of the Workstation Properties window or increase Rate limit sampling interval in the Carrier Security page of the Global Properties window. Also, it is possible to change the drop and alert behavior of the rate limiting feature by editing the <code>gtp_rate_limit_drop</code> and <code>gtp_rate_limit_alert</code> properties using the GUI Dbedit tool. |
| GTP: T-PDU is a GTP message                             | This T-PDU packet (The internal packet of a G-PDU) is a GTP packet by itself. This may indicate on attempt to inject GTP packets into the system.                               | If you do want to enable such type of packets, you can check the Allow GTP in GTP in the Carrier Security page of the Global Properties tab (equivalent to setting <code>block_gtp_in_gtp</code> to 0).                                                                                                                                                                                                                                                                                                                                                                               |
| GTP: Invalid End User IP Address                        | This T-PDU packet (The internal packet of a G-PDU) has an end user address IP that does not match the end user IP address of the PDP context associated with this G-PDU packet. | Uncheck the Enforce GTP AntiSpoofing property in the Carrier Security page of the Global Properties window (this is equivalent to setting the <code>gtp_anti_spoofing</code> property to 0). To uncheck only GTP IPv6 AntiSpoofing, set the <code>gtp_ipv6_anti_spoofing</code> property to 0.                                                                                                                                                                                                                                                                                        |
| GTP intra-tunnel Inspection: Forbidden MS-to-MS traffic | The end user address of this T-PDU does not conform to the end user Domain Policy defined for the APN of the PDP Context associated with this G-PDU packet.                     | Change the end user Domain Policy in the APN Properties window.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Log Message                    | Meaning                                                                                                                                                                                                                       | Resolution                                                                  |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Illegal Handover               | An Update Request was initiated from a new SGSN (source IP) which is not in the Handover group of the old SGSN of the tunnel. You can see the new SGSN IP in the Source column and the old SGSN IP in the SGSN Signal column. | Adjust the GSN Handover Group definitions in the GSN Handover Group window. |
| Illegal Handover GSN Signaling | Illegal redirection attempt for GSN signaling. The GSN Signaling Information Element IP is not in the same Handover group as the Source IP of the message. You can see both IPs in the log.                                   | Adjust the GSN Handover Group definitions in the GSN Handover Group window. |
| Illegal Handover - GSN Traffic | Illegal redirection attempt for GSN traffic. The GSN traffic Information Element IP is not in the same Handover group as the source IP of the message. You can see both IPs in the log.                                       | Adjust the GSN Handover Group definitions in the GSN Handover Group window. |

| Log Message                        | Meaning                                                                                                                                                                              | Resolution                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Illegal Handover Recreate PDPC     | This "Create PDP Context Request" PDU did not conform to the handover policy.                                                                                                        | If the <code>gtp_allow_recreate_pdpc</code> property is set to <code>open</code> , the policy allows recreating a tunnel using SGSN addresses complying with the SGSN handover policy.<br>GSN handover and GSN redirection are only allowed within a GSN Handover Group.<br>If this PDU does conform to your handover policy, adjust the GSN Handover Group definitions in the GSN Handover Group window.                                      |
| Illegal response cause             | The response cause in this response message is illegal for this message type.                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Invalid G-PDU                      | Relevant for V0 G-PDUs. The SGSN IP, GGSN IP or Flow Label of the G-PDU does not match the definitions of the tunnel the G-PDU belongs to. (Tunnel association is according to TID). | You can remove flow label compliance on the Carrier Security page of the Global Properties window. However if the Flow Labels are wrong, it is recommended to investigate the cause. IP checking cannot be disabled.                                                                                                                                                                                                                           |
| Invalid Signaling Recreate Req PDU | Relevant for V0. There was an attempt to create a PDP Context of an already established tunnel.                                                                                      | The recreate policy of established tunnels is determined by the <code>gtp_allow_recreate_pdpc</code> property.<br>A <code>strict</code> policy allows recreating a tunnel using only the identical GSN addresses. If a tunnel is recreated using different GSN addresses and we are in a strict "Re-Create" Policy - the create is dropped and this message is logged. An <code>open</code> policy allows GSN handover for tunnel recreations. |

| Log Message                                    | Meaning                                                                                                                                                                                                           | Resolution                                                                                                                                                                    |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Invalid Signaling Req PDU                      | Relevant for V0 Delete Request, V0 Update Request, and V0->V1 Update Request. Either the source IP address, dest. IP address, or flow label does not match those of the tunnel (TID) to which the packet belongs. | Flow label verification can be disabled by deselecting the Verify Flow Label setting, found in the Carrier Security tab of Global Properties. IP checking cannot be disabled. |
| Invalid Signaling Flow Label PDU (Update Resp) | V0 Update Resp. The flow label does not match the tunnel (TID) to which the packet belongs.                                                                                                                       | Flow label verification can be disabled by deselecting the Verify Flow Label setting, found in the Carrier Security tab of Global Properties. IP checking cannot be disabled. |
| Invalid Signaling Flow Label PDU (Create Resp) | V0 Create Resp. The flow label does not match the tunnel (TID) to which the packet belongs.                                                                                                                       | Flow label verification can be disabled by deselecting the Verify Flow Label setting, found in the Carrier Security tab of Global Properties. IP checking cannot be disabled. |
| Invalid Signaling Flow Label PDU (Delete Resp) | V0 Delete Resp. The flow label does not match the tunnel (TID) to which the packet belongs.                                                                                                                       | Flow label verification can be disabled by deselecting the Verify Flow Label setting, found in the Carrier Security tab of Global Properties. IP checking cannot be disabled. |
| IP is not in the APN domain                    | The assigned static or dynamic end user IP is not part of end user Domain defined for the related APN.                                                                                                            | This packet is dropped according to the APN end user Domain defined in SmartConsole.                                                                                          |
| Malformed Path Management PDU                  | This Path management PDU does not conform to GTP standards.                                                                                                                                                       | Path management PDUs are verified against GTP Release 1997 and 1999 Standards.                                                                                                |

| Log Message                                                 | Meaning                                                                                                                                                                           | Resolution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No Match on Create PDP Context PDU                          | A "Create PDP Context Request" PDU was not matched on the Rule Base.                                                                                                              | <p>The allowed types of "Create PDP Context Request" PDUs are defined in the Rule Base using Source, Destination and the Advanced GTP Service Properties window.</p> <p>If the combination of the above in the dropped PDU should have been allowed, please review your Rule Base to allow this traffic.</p> <p>If the last rule in the Security Policy Rule Base is an "Accept" rule, set Produce extended log on unmatched PDUs to "Last" instead of "Before Last" in the Carrier Security page of the Global Properties window.</p> |
| Out of range sequence number                                | This G-PDU carries an out-of-range sequence number.                                                                                                                               | <p>Enforcement of G-PDU sequence numbers is determined in the Carrier Security page of the Global Properties window, where you can also define the maximum allowed deviation for all Carrier Security Gateways.</p> <p>Also, it is possible to change the drop and alert behavior of the rate limiting feature by editing <code>gtp_sequence_deviation_drop</code> and <code>gtp_sequence_deviation_alert</code> properties using the GUI Dbedit tool.</p>                                                                             |
| Packet or some Information Element is shorter than expected | During stateful inspection, this packet (PDU) was shorter than expected.                                                                                                          | This packet does not have the minimum length to hold the GTP header information, or the packet size is small than indicated by the length field in the GTP header.                                                                                                                                                                                                                                                                                                                                                                     |
| Passed maximum create request                               | Too many re-transmissions of the same delete request were received (while create response not received yet by the Carrier Security Gateway). This request packet will be dropped. | Set the <code>gtp_max_req_retransmit</code> variable to the number of allowed outstanding re-transmits.                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Log Message                    | Meaning                                                                                                                                                                           | Resolution                                                                                                                                                                                                                   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Passed maximum delete request  | Too many re-transmissions of the same delete request were received (while delete response not received yet by the Carrier Security Gateway). This request packet will be dropped. | This can occur if the Carrier Security Gateway is configured to close all end user connections using the SAM API.<br>Set the <code>gtp_max_req_retransmit</code> variable to the number of allowed outstanding re-transmits. |
| Passed maximum update request  | Too many re-transmissions of the same update request were received (while update response not received yet by the Carrier Security Gateway). This request packet will be dropped. | Set the <code>gtp_max_req_retransmit</code> variable to the number of allowed outstanding re-transmits.                                                                                                                      |
| re-using TEID Control Downlink | The specified TEID of this tunnel create attempt is in use by another tunnel (for the same SGSN-GGSN pair). The new tunnel is created.                                            | If the attribute <code>gtp_allow_recreate_pdpc</code> is set to <code>strict</code> , the new tunnel is not created in this case.                                                                                            |
| re-using TEID Data Downlink    | The specified TEID of this tunnel create attempt is in use by another tunnel (for the same SGSN-GGSN pair). The new tunnel is created.                                            | If the attribute <code>gtp_allow_recreate_pdpc</code> is set to <code>strict</code> , the new tunnel is not created in this case.                                                                                            |

| Log Message                               | Meaning                                                                                                                                | Resolution                                                                                                                                               |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| re-using TEID<br>Data Uplink              | The specified TEID of this tunnel create attempt is in use by another tunnel (for the same SGSN-GGSN pair). The new tunnel is created. | If the attribute <code>gtp_allow_recreate_pdpc</code> is set to <code>strict</code> , the new tunnel is not created in this case.                        |
| re-using TEID<br>Control Uplink           | The specified TEID of this tunnel create attempt is in use by another tunnel (for the same SGSN-GGSN pair). The new tunnel is created. | If the attribute <code>gtp_allow_recreate_pdpc</code> is set to <code>strict</code> , the new tunnel is not created in this case.                        |
| re-using TEID<br>Control Uplink,<br>SRC=0 | The specified TEID of this tunnel create attempt is in use by another tunnel (for the same SGSN-GGSN pair). The new tunnel is created. | If the attribute <code>gtp_allow_recreate_pdpc</code> is set to <code>strict</code> , the new tunnel is not created in this case.                        |
| Request/<br>Response<br>Mismatch          | This Signaling Response PDU carries a wrong Sequence number or does not match any Signaling Request.                                   | Signaling Response PDUs must match a previously approved Signaling Request PDU in order to pass the Carrier Security Gateway. This cannot be configured. |

| Log Message                                | Meaning                                                                                                                                                                                                                                                                                           | Resolution                                                                                                                                                                                                                                            |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TEID 0 not allowed for Update message type | A V1 Update Request has TEID 0. This is valid only for V0 to V1 handover cases. However the imsi and nsapi Information Elements of this Request do not match an existing V0 tunnel.                                                                                                               | This packet will be dropped.                                                                                                                                                                                                                          |
| TID 0 not allowed for this message type    | A Signaling PDU carries a NULL TID violating the GTP protocol.                                                                                                                                                                                                                                    | This packet violated basic packet integrity and will not pass through the Carrier Security Gateway.                                                                                                                                                   |
| Unestablished Tunnel                       | <p>This signaling or data packet belongs to an unestablished tunnel.</p> <ul style="list-style-type: none"> <li>■ For V0, the packet has a Tunnel ID (TID) of an Unknown PDP Context.</li> <li>■ For V1, the packet has a Tunnel EndPoint Identifier (TEID) of an Unknown PDP Context.</li> </ul> | PDUs can only pass the Carrier Security Gateway if they carry a Tunnel ID (V0) or a Tunnel EndPoint ID (V1) of a previously established PDP context that was not yet terminated. This packet violates basic tunnel integrity and will not be allowed. |
| Unknown GTP Message Type                   | This packet constitutes an unsupported GTP Signaling message.                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                       |

| Log Message         | Meaning                                                     | Resolution                  |
|---------------------|-------------------------------------------------------------|-----------------------------|
| Unsupported version | A GTP packet with version other than V0 or V1 was received. | The packet will be dropped. |

## Adding Information Elements to Logs

Carrier Security 6.0 provides the option of including certain Information Elements to logs with GTP information. These Information Elements are:

- RAT - (Radio Access Type)
- IMEI-SV (International Mobile Equipment Identity - Software Version)
- MS-Time Zone
- Mobile User Location

To add these Information Elements to the log, use the GuiDBedit database tool to set the attribute `gtp_log_additional_fields` to `true`. The default setting is `false`. Adding Information Elements to Logs

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# Advanced Configurations

This section describes Carrier Security advanced configurations.

## GRX Redundant Deployment

Cellular operators strive to maintain 99.999% network availability. In this effort, they sometimes use two separate connections to the GRX (GPRS Roaming eXchange) network using two separate GRX providers on different sites.

This is somewhat similar to a dual ISP scenario, but with the difference that each GRX connection is set up in a different physical site protected by its own Carrier Security Gateway. The demanded availability requires all UMTS/LTE connections to survive in the event that one of these GRX connections will fail. For that reason, the cellular operator needs to synchronize the kernel state tables between the two Carrier Security gateways which are situated on physically distant locations. This requires the synchronization traffic to be transported over a wide area network (WAN). Synchronization traffic originally was designed to pass between cluster members only on a Local Area Network (LAN). Here we present a method that has been validated to work well to synchronize the Carrier Security cluster members over a Wide Area Network, and maintain full uptime and security for all PDP contexts during a "GRX failover". Uptime and security is maintained for PDP Contexts that existed before the failover occurred, and for the new ones that were initiated after the failover occurred. All PDP Contexts are maintained and secured during the failback as well.

The basic idea behind the solution presented here is to deploy a Layer 2 tunnel between the two Carrier Security Gateways, which allows the Carrier Security cluster members to operate normally, as if they were on the same LAN, even though the Layer 2 tunnel traffic is being routed over a WAN network.

## Asymmetric Routing

This solution works for both symmetric and asymmetric routing. Asymmetric routing takes place when some of the packets of a certain PDP session pass through one GRX, while other packets of the same PDP Context pass through another GRX. This can take place in either direction, i.e., to or from the partner.

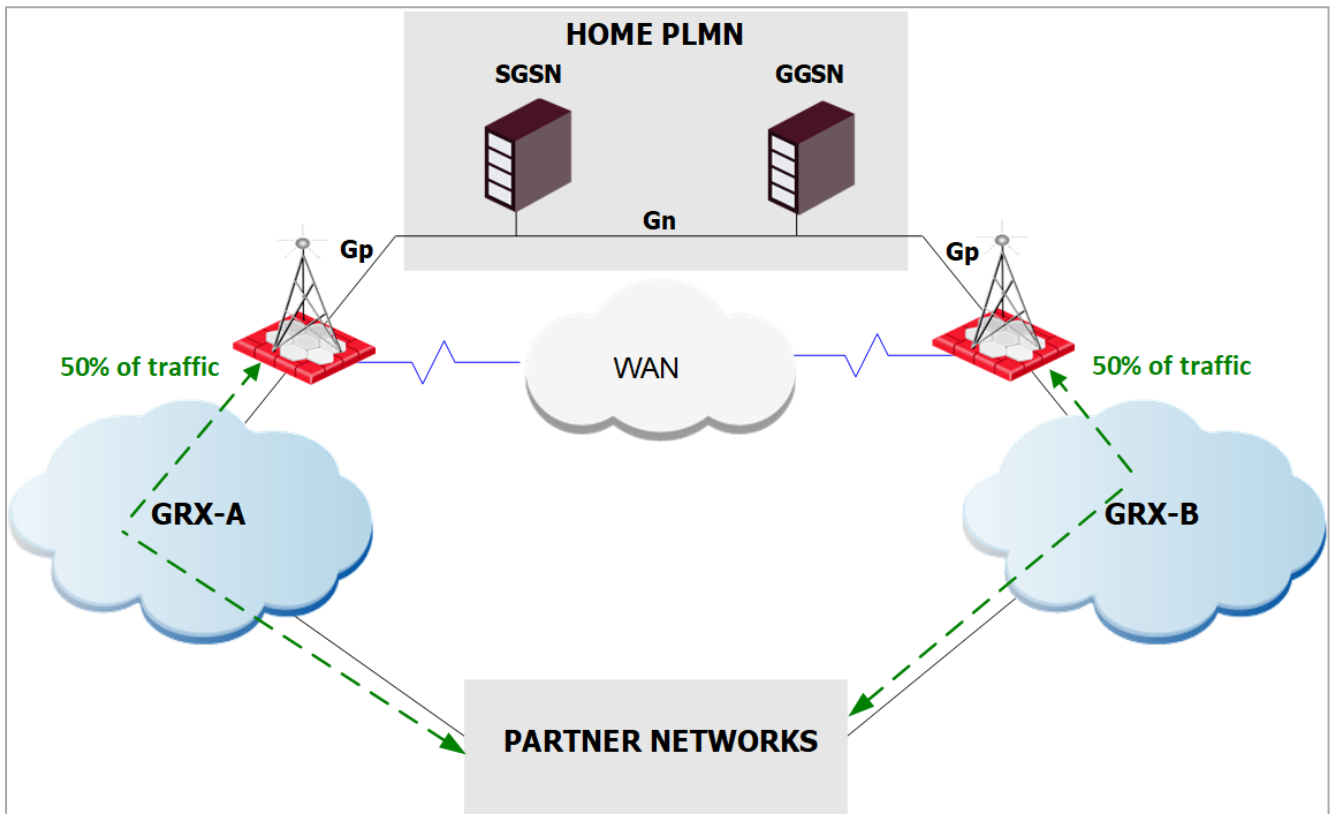
In this deployment, asymmetric routing can be manifested in a few ways:

- A GTP Create Request passes through GRX-A, and the corresponding GTP Create Response returns through GRX-B.
- Same as #1 for Update, Delete, Echo exchanges.
- T-PDU traffic may be split between GRX-A and GRX-B, in both directions (to and from the partner).

Asymmetric routing is supported by holding critical packets at the receiving Gateway until the peer gateway has acknowledged that its information on these packets is in sync. This is true, for example, for all Request type messages, since the peer Gateway must register a Request packet before the corresponding Response message arrives.

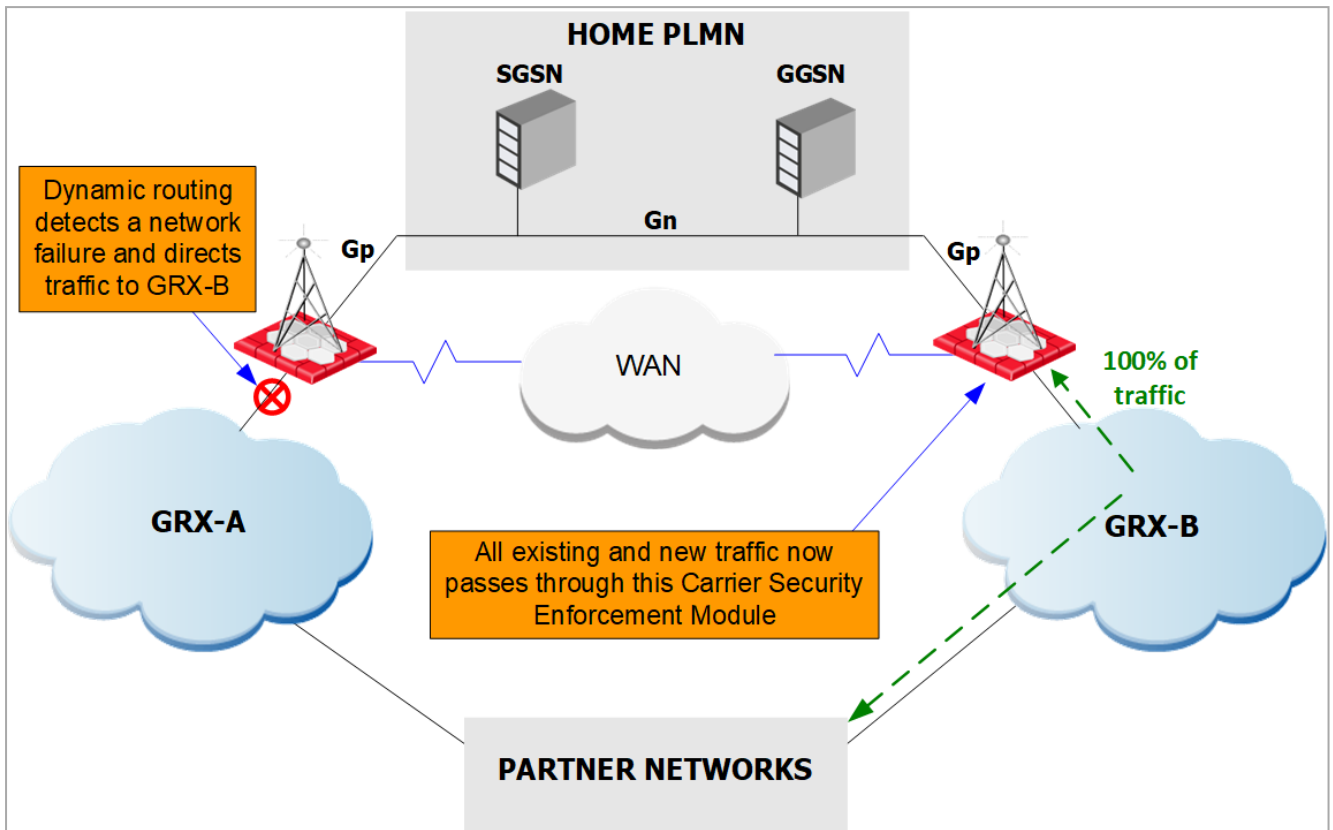
During normal operation, traffic is load-shared between the two GRXs, and consequently load-shared between the two Carrier Security Gateways. The traffic flow is according to the operator routing settings.

#### GRX load sharing during normal operation:



If any point on the network of one of the GRXs should fail, all traffic takes the path of the second, fully-functional GRX.

### Traffic failover during GRX failure:



The path change occurs via dynamic routing settings using OSPF, BGP, etc. The data remains synchronized between the two Gateways.

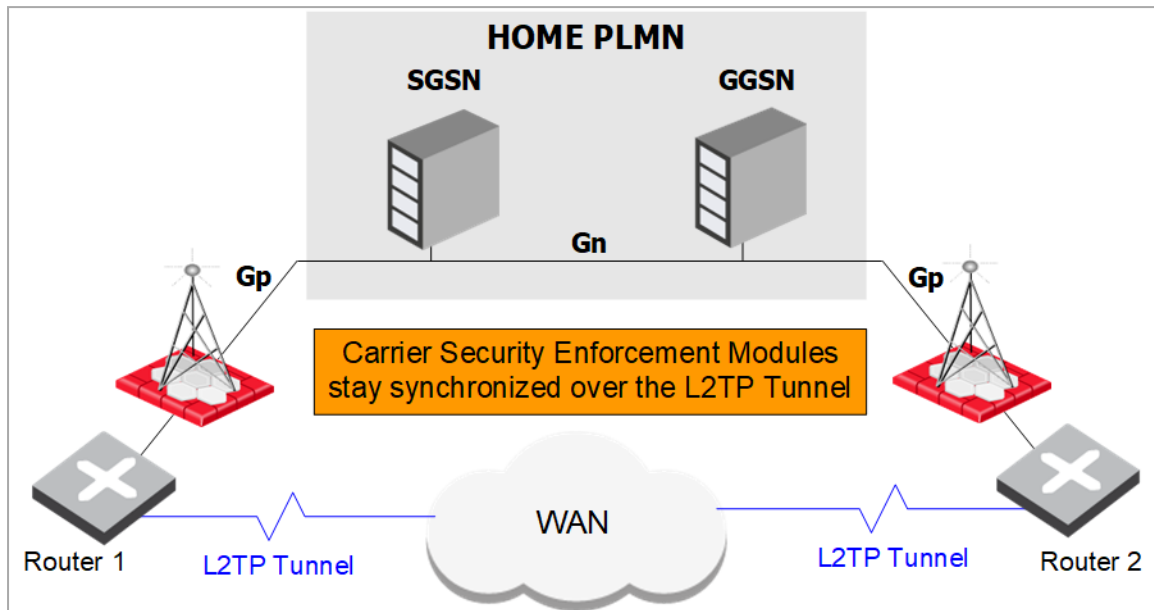
## Configuring a GRX Redundant Deployment

The distributed Carrier Security cluster consists of two Carrier Security Gateways.

**Note** - It is likely that more than two Carrier Security Gateways in such a deployment will work as well (e.g., in a deployment with three GRX connections), although this has not been verified.

1. Define a Carrier Security cluster, using 3rd Party HA mode, and add cluster members.
2. Set up the cluster to support non sticky connections, which enables the proper handling of asymmetric routing flows.
3. Set up a Layer 2 tunnel, such as L2TP, GRE, etc., between the two Carrier Security Gateways. The interfaces on both Gateways constituting the sync network should connect to the Layer 2 tunneling device on a LAN (or crossover cable).

## Synchronizing Carrier Security over a WAN:



- If desired, and the Layer 2 tunneling device supports it, establish encryption on the Layer 2 tunnel.
- On the interfaces of the sync network, set the MTU to 1400. A value higher than 1400 may cause PMTU discovery procedures not supported by the tunneling device.

## Testing the Setup

Perform the tests below to verify that the synchronization over the WAN network is operating properly.

1. Check on each Carrier Security member that it can see the other member over Layer 2. This is done by running the command `arp -a` on one member, and verifying that it displays the MAC address of the other member. This will indicate that the Layer 2 tunnel is set up correctly.
2. Using tools such as `tcpdump`, `fw monitor`, etc., verify that sync traffic is being transmitted on the sync network.
3. Verify the health of the sync mechanism by checking the output of the command `cphaprob stat` and `cphaprob [-reset] syncstat`. Refer to the ClusterXL User Guide included on the CD for details on the output of these commands.
4. Verify that the GTP kernel tables are synchronized between the 2 members after passing some GTP traffic between them. Use the command `fw tab -s | grep gtp` and compare the results. In general both members should show the same values for the tables. Slight differences are acceptable.

5. Simulate failover by either shutting down a gateway and/or diverting all traffic to the other GRX path. In all cases, existing and new PDP contexts should remain intact.
6. Perform a reboot of Carrier Security member and verify that existing and new PDP Contexts remain intact.
7. Perform tests of asymmetric routing by setting up such routes in your network.

## Fine-tuning Cluster Synchronization Parameters

For suggestions for fine tuning synchronization parameters, see the [R82 ClusterXL Administration Guide](#).

## Stripping Information Elements

In situations where an operator has advanced GSNs, using up-to-date or preliminary Information Elements, it may occur that a partner of this operator will have a GTP-aware firewall that drops these messages. This is due to the "suspicious" nature of firewalls to "unknown" traffic.

Carrier Security can be configured to strip specific Information Elements for GTP messages exchanged with specific partners in order to avoid cases where the firewall of the partner will drop the message. Today this already has potential relevancy for the following Information Elements:

- IMEI-SV
- User Location Information
- RAT
- Time Zone

To strip Information Elements from traffic destined for specific partners, do the following:

1. Create a new GTP v2 service as detailed in **Enforcing a More Granular GTP Security Policy** and select Save.
2. Open the **GuiDBedit** tool. Go to Services > services, and select the new service.
3. In the variables list, select the field `stripped_information_elements` and insert the numeric values of the Information Elements you want to remove. The delimiter can be either ',' (comma) or ' ' (space). Valid values for the Information Elements are from 1-30 (TV Information Elements), 116-127 (TV reserved Information Elements - they are all reserved except 127 that is in use), and 128 - 255 (TLV Information Elements).

For example, if you want to strip the RAT and IMEI Information Elements enter: 151 , 154

**Note** - If the input is illegal, the policy will not install.

1. Save and close **GuiDBedit**.
2. Add the new service to the Security Rule Base for the partner(s) for whom you want to strip the IEs, and install policy. The selected Information Elements will be removed from traffic to the selected partner(s).

## Capacity Management

This section covers Automatic Tunnel Capacity and GTP Tunnel Aggressive aging.

### GTP Tunnel Capacity

Tunnel capacity is configured per gateway object in the Carrier Security pane of the gateway object properties. By default, tunnel capacity is set to *automatic*.

### GTP Tunnels Aggressive Aging

Aggressive aging for GTP tables monitors the gateway for two main indicators to determine the system's capability to accept new GTP tunnels:

- Amount of GTP tunnels already established in the *gtp\_tunnels* table
- Free system memory

When the gateway detects one of the indicators exceeding the upper thresholds, the system will activate aggressive aging. While active, the gateway will attempt to delete idle GTP tunnels from the *gtp\_tunnels* table before recording new tunnels.

### Fine Tuning

While the default values should fit most scenarios and deployments, you can change them to address conditions such as an expected high load on a roaming gateway. These default values can be changed in the Global Properties window.

| Parameter          | Default / Min / Max value | Description                                                                       |
|--------------------|---------------------------|-----------------------------------------------------------------------------------|
| Aggressive aging   | 0                         | Enables Aggressive Aging                                                          |
| Aggressive Timeout | 3600                      | Duration in seconds before a tunnel is considered idle and eligible for deletion. |

| Parameter                      | Default / Min / Max value | Description                                                                                                                                                                           |
|--------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tunnel activation threshold    | 80 / 0 / 100              | Upper thresholds for activating aggressive aging. When tunnel amount or memory usage exceeds theses values (as a percent from the respective limits) - aggressive aging is activated. |
| Memory activation threshold    | 80 / 0 / 100              |                                                                                                                                                                                       |
| Tunnel de-activation threshold | 60 / 0 / 100              | Lower thresholds for deactivating aggressive aging. When tunnel amount or memory usage drop below theses values aggressive aging is deactivated.                                      |
| Memory de-activation threshold | 60 / 0 / 100              |                                                                                                                                                                                       |

**Note** - These guidelines are enforced and verified by SmartConsole before a policy is pushed to the gateway:

- **Thresholds**

The upper threshold must be lower than a hundred and greater than the lower threshold which must be greater than a zero.

If a threshold is not within the permitted range, both upper and lower thresholds for the category (tunnel count or memory) are reset to the default values.

- **Aggressive time-out**

The *gtp\_tunnels* table time out must be greater than the *gtp\_aggr* timeout which must be greater than the *cphwd\_gtp\_refresh\_interval* x 1.5

If the aggressive timeout is not within the range it will be corrected automatically to a valid value.

## SCTP

SCTP offers greater reliability over other transport layer protocols (such as UDP and TCP) by supporting multiple IP paths (multihoming) to a peer endpoint. For each multihomed SCTP endpoint, multiple IP addresses serve each SCTP association.

**Note** - You can create one rule for an SCTP connection, but for each multihomed endpoint you must create a group of IP addresses associated with the connection.

In case of failover, each new active connection is checked for access policy and state. A new active connection is not automatically accepted because of an existing association.

## SCTP Supported Features

SCTP fully complies with RFC4960, which describes the Stream Control Transmission Protocol, and supports these features:

- SCTP stateless verification - makes sure that each SCTP packet complies with RFC 4960 regardless of the packet's state.
- SCTP stateful inspection - accepts SCTP packets according to the association state.
- Support for multihomed SCTP - opens multiple child connections under the established association.
- SCTP packet filtering - filters for access control through policy rule definition.
- SCTP with static NAT.
- SCTP Acceleration.

## Configuring SCTP Inspection

When a Carrier license is installed, you can specify SCTP services in your Firewall rules. SCTP Inspection occurs in these cases:

- There is a match on a rule containing an SCTP or Diameter SCTP service in the **Service** cell.
- There is a match on a rule with **Service= Any** and this SCTP service has **Match for any** selected.

**To activate SCTP Inspection:**

1. Open **SmartConsole**.
2. In **Objects > Object Explorer** click **New > Service > SCTP**.

On the **General** page configure:

- **Name** - The name of the service. The name assigned here must be the same as the server service name (as in the services file). If NIS is used, the firewall automatically retrieves the information from NIS.
  - **Port** - The number of the port that gives this service.
3. Click **Advanced** and configure:

The **Advanced SCTP Service Properties** window opens.

- **Source Port** - Port number for the client side service. If specified, only those Source port Numbers will be Accepted, Dropped, or Rejected during packet inspection. Otherwise, the source port is not inspected.
- **Keep connections open after policy has been installed** - If the connections are not allowed in the new policy, they are still kept. This overrides the settings in the Connection Persistence page. If you change this property, the change does not have effect on open connections, but only future connections.
- **Enable Aggressive Aging** - Sets short (aggressive) timeouts for idle connections. When a connection is idle for more than its aggressive timeout value, it is marked as `eligible for deletion`. When memory consumption or connections table capacity exceeds a user-defined threshold (high watermark), aggressive aging starts. Each incoming connection starts to delete `k` (10 by default) connections that are eligible for deletion. This continues until memory consumption or connections capacity decreases below the low value.
- **Synchronize connections if State Synchronization is enabled on cluster** - Enables state-synchronized High Availability or Load Sharing on a ClusterXL or OPSEC-certified cluster. Of the services allowed by the Rule Base, only those with **Synchronize connections on cluster** selected are synchronized as they go through the cluster. By default, all new and existing services are synchronized.

4. Click OK.

5. Open **Global properties > Stateful Inspection**.

Configure these **Stateful Inspection** options:

| Option               | Meaning                                                                                                                                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCTP start timeout   | <ul style="list-style-type: none"> <li>■ An SCTP connection times out if the interval between the arrival of the first packet and establishment of the connection (STCP four-way handshake) exceeds the SCTP start timeout in seconds.</li> <li>■ Attribute name in GuiDBedit: <code>sctpstarttimeout</code></li> </ul> |
| SCTP session timeout | <ul style="list-style-type: none"> <li>■ Length of time an idle connection remains in the Security Gateway connections table.</li> <li>■ Attribute name in GuiDBedit: <code>sctptimeout</code></li> </ul>                                                                                                               |
| SCTP end timeout     | <ul style="list-style-type: none"> <li>■ A SCTP connection will only terminate SCTP end timeout seconds after two FIN packets (one in each direction: client-to-server, and server-to-client) or an RST packet.</li> <li>■ Attribute name in GuiDBedit: <code>sctpendtimeout</code></li> </ul>                          |

Configure these options for **Out of state packets**:

| Option                         | Meaning                                                                                                                                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drop out of state Sctp packets | <ul style="list-style-type: none"> <li>Drop Sctp packets that are not consistent with the current state of the Sctp connection.</li> <li>Attribute name in GuiDBedit: fw_drop_out_of_state_sctp</li> </ul> |
| Log on drop                    | <ul style="list-style-type: none"> <li>Generates a log entry when out of state Sctp packets are dropped.</li> <li>Attribute name in GuiDBedit: fw_log_out_of_state_sctp</li> </ul>                         |

#### To deactivate out of state packet drop with SmartConsole:

1. In SmartConsole, go to **Global properties > Stateful Inspection**.
2. Clear the **Drop out of state Sctp packets** option.
3. Save and install the policy.

#### To deactivate packet inspection with GuiDBedit:

1. Open **GuiDBedit**.
2. Search for: fw\_sctp\_packet\_inspection.
3. Set the property to **false**.
4. Save the database and install policy.

## Configuring Sctp Acceleration

To enable Sctp acceleration, run this command on the Security Gateway or cluster member:

```
sim feature sctp on
```

To disable Sctp acceleration, run this command on the Security Gateway or cluster member:

```
sim feature sctp off
```

**Note** -If Sctp acceleration is activated and Sctp inspection is deactivated, the Performance Pack accelerates all Sctp packet types.

## Configuring Sctp NAT

Sctp NAT overrides the currently defined NAT policy. When this feature is not activated, Sctp connections do not use NAT.

To activate Sctp NAT, run this command on the Security Gateway:

```
fw ctl set int fwx_enable_sctp_nat 1
```

To deactivate Sctp NAT, run this command on the Security Gateway:

```
fw ctl set int fwx_enable_sctp_nat 0
```

# Glossary

## A

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### AA

Anonymous Access - the network does not know the real identity of the mobile, opposite of non-anonymous access.

### Anti-Bot

Check Point Software Blade on a Security Gateway that blocks botnet behavior and communication to Command and Control (C&C) centers. Acronyms: AB, ABOT.

### Anti-Spam

Check Point Software Blade on a Security Gateway that provides comprehensive protection for email inspection. Synonym: Anti-Spam & Email Security. Acronyms: AS, ASPAM.

### Anti-Virus

Check Point Software Blade on a Security Gateway that uses real-time virus signatures and anomaly-based protections from ThreatCloud to detect and block malware at the Security Gateway before users are affected. Acronym: AV.

### AP

Access Point - entry point to an external network.

### APN

Access Point Name - the identifier of an external packet data network.

### Application Control

Check Point Software Blade on a Security Gateway that allows granular control over specific web-enabled applications by using deep packet inspection. Acronym: APPI.

### Audit Log

Log that contains administrator actions on a Management Server (login and logout, creation or modification of an object, installation of a policy, and so on).

**B**

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**Bearer**

A service that allows transmission of information signals between network interfaces. The bearer or data service is used to provide the same level of packet-forwarding treatment for user data as it travels across the network.

**BG**

Border Gateway - a logical box that connects two (or more) operators together via Inter-PLMN backbone; protects operator's intra-PLMN network against intruders.

**Bridge Mode**

Security Gateway or Virtual System that works as a Layer 2 bridge device for easy deployment in an existing topology.

**BSSAP+**

Base Station System Application Part+ - the protocol between SGSN and MSC/VLR

**BSSGP**

Base Station System GPRS Protocol - the protocol between SGSN and BSS.

**C**

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**CCU**

Channel Codec Unit - the functional element in BSS that handles low level GPRS control in radio.

**CLNS**

Connection Less Network Service; similar to the IP protocol.

**Cluster**

Two or more Security Gateways that work together in a redundant configuration - High Availability, or Load Sharing.

**Cluster Member**

Security Gateway that is part of a cluster.

**Compliance**

Check Point Software Blade on a Management Server to view and apply the Security Best Practices to the managed Security Gateways. This Software Blade includes a library of Check Point-defined Security Best Practices to use as a baseline for good Security Gateway and Policy configuration.

**CONS**

Connection Oriented Network Service, similar to the X.25 protocol.

**Content Awareness**

Check Point Software Blade on a Security Gateway that provides data visibility and enforcement. Acronym: CTNT.

**CoreXL**

Performance-enhancing technology for Security Gateways on multi-core processing platforms. Multiple Check Point Firewall instances are running in parallel on multiple CPU cores.

**CoreXL Firewall Instance**

On a Security Gateway with CoreXL enabled, the Firewall kernel is copied multiple times. Each replicated copy, or firewall instance, runs on one processing CPU core. These firewall instances handle traffic at the same time, and each firewall instance is a complete and independent firewall inspection kernel. Synonym: CoreXL FW Instance.

**CoreXL SND**

Secure Network Distributer. Part of CoreXL that is responsible for: Processing incoming traffic from the network interfaces; Securely accelerating authorized packets (if SecureXL is enabled); Distributing non-accelerated packets between Firewall kernel instances (SND maintains global dispatching table, which maps connections that were assigned to CoreXL Firewall instances). Traffic distribution between CoreXL Firewall instances is statically based on Source IP addresses, Destination IP addresses, and the IP 'Protocol' type. The CoreXL SND does not really "touch" packets. The decision to stick to a particular FWK daemon is done at the first packet of connection on a very high level, before anything else. Depending on the SecureXL settings, and in most of the cases, the SecureXL can be offloading decryption calculations. However, in some other cases, such as with Route-Based VPN, it is done by FWK daemon.

**CPUSE**

Check Point Upgrade Service Engine for Gaia Operating System. With CPUSE, you can automatically update Check Point products for the Gaia OS, and the Gaia OS itself.

**CS**

Circuit Switched; opposite of packet switched.

**CSCF**

Call Session Control Function. A set of roles for SIP servers or proxies that handle SIP signal packets in the IP Multimedia Subsystem (IMS).

**D**

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**DAIP Gateway**

Dynamically Assigned IP (DAIP) Security Gateway is a Security Gateway, on which the IP address of the external interface is assigned dynamically by the ISP.

**Data Loss Prevention**

Check Point Software Blade on a Security Gateway that detects and prevents the unauthorized transmission of confidential information outside the organization. Acronym: DLP.

**Data Service**

A service that allows transmission of information signals between network interfaces. The bearer or data service is used to provide the same level of packet-forwarding treatment for user data as it travels across the network.

**Data Type**

Classification of data in a Check Point Security Policy for the Content Awareness Software Blade.

**Diameter**

An authentication, authorization and accounting protocol that has many features not included in the legacy RADIUS protocol.

**Distributed Deployment**

Configuration in which the Check Point Security Gateway and the Security Management Server products are installed on different computers.

**DRX**

Discontinuous Reception - when MS receives intermittently.

**Dynamic Object**

Special object type, whose IP address is not known in advance. The Security Gateway resolves the IP address of this object in real time.

**E**

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**EDGE**

Enhanced Data-rates for GSM Evolution, a technology for enhancing GSM to deliver mobile data and multimedia services; an alternative to UTMS.

**End-to-End Security**

A single encrypted and authenticated tunnel through the operator network, reaching from the wireless device to the server. End-to-end security requires that the entire connection be IP-based; this can occur only in third-generation networks.

**Endpoint Policy Management**

Check Point Software Blade on a Management Server to manage an on-premises Harmony Endpoint Security environment.

**Expert Mode**

The name of the elevated command line shell that gives full system root permissions in the Check Point Gaia operating system.

**G**

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**G-PDU**

A user data message, comprising a G-PDU and a GTP header.

**Gaia**

Check Point security operating system that combines the strengths of both SecurePlatform and IPSO operating systems.

**Gaia Clish**

The name of the default command line shell in Check Point Gaia operating system. This is a restricted shell (role-based administration controls the number of commands available in the shell).

**Gaia Portal**

Web interface for the Check Point Gaia operating system.

**Gb**

Interface between an SGSN and a BSS.

**Gc**

Interface between a GGSN and an HLR.

**Gd**

Interface between a SMS-GMSC and an SGSN, and between a SMS-IW MSC and an SGSN.

**Gf**

Interface between an SGSN and an EIR.

**GGSN**

Gateway GSN (GPRS Support Node).

**Gi**

Reference point between GPRS and an external packet data network.

**GMM/SM**

GPRS Mobility Management and Session Management - protocol stack between MS and SGSN that handles GPRS attach/detach, PDP context activation/deactivation, etc.

**Gn**

Interface between two GSNs within the same PLMN.

**Gp**

Interface between two GSNs in different PLMNs. The Gp interface allows support of GPRS network services across areas served by the co-operating GPRS PLMNs.

**GPRS**

General Packet Radio System, a non-voice value-added service for faster data transactions over a mobile telephone network, designed for deployment on GSM and TDMA-based mobile networks. GPRS overlays a packet-based air interface on the existing switched network.

**Gr**

Interface between an SGSN and an HLR.

**Gs**

Interface between an SGSN and an MSC/VLR.

**GSM**

Global System for Mobile Communications (originally Groupe Speciale Mobile, hence the acronym) - a second generation time-division mobile network standard.

**GSN**

GPRS Support Node.

**GTP**

GPRS Tunnel Protocol.

**GTP Tunnel**

In GTP version 0 GTP tunnel is defined by two associated PDP Contexts in different GSN nodes and is identified with a Tunnel ID. (1) In GTP version 1/2, a GTP tunnel in the GTP-C plane is defined for all PDP Contexts/sessions with the same PDP address and APN (for Tunnel Management messages), or for each MS (for messages not related to Tunnel Management). A GTP tunnel is identified in each node with a TEID, an IP address and a UDP port number. (2) In GTPv1 GTP tunnel in the GTP-U plane is defined for each PDP Context in the GSNs. While in GTPv2 a bearer is used. (3) In GTP version 2, a GTP-C tunnel is defined for all PDP sessions with same PDP address and TEID, For GTP-U plane traffic a Bearer is created. (4) In all versions, a GTP tunnel is necessary to forward packets between an external packet data network and an MS user.

**H**

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**HLR**

Home location register - a central database that contains user-related and subscription-related information.

**Hotfix**

Software package installed on top of the current software version to fix a wrong or undesired behavior, and to add a new behavior.

**HPLMN**

Home Public Land Mobile Network - the home network.

**HSCSD**

High Speed Circuit Switched Data - a new GSM service for circuit switched connections.

**HSPA**

High Speed Packet Access. An improved third generation mobile communication protocol that significantly enhances data transfer. It is a combination of two protocols: (1) HSUPA - High Speed Uplink Packet Access (2) HSDPA - High Speed Downlink Packet Access.

**HTTPS Inspection**

Feature on a Security Gateway that inspects traffic encrypted by the Secure Sockets Layer (SSL) protocol for malware or suspicious patterns. Synonym: SSL Inspection. Acronyms: HTTPSI, HTTPSi.

**I**

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**ICA**

Internal Certificate Authority. A component on Check Point Management Server that issues certificates for authentication.

**Identity Awareness**

Check Point Software Blade on a Security Gateway that enforces network access and audits data based on network location, the identity of the user, and the identity of the computer. Acronym: IDA.

**Identity Logging**

Check Point Software Blade on a Management Server to view Identity Logs from the managed Security Gateways with enabled Identity Awareness Software Blade.

**IE**

Information Element - a group of information which may be included within a signaling message or data flow.

**IETF**

Internet Engineering Task Force - Internet standardization organization.

**IMSI**

International Mobile Subscriber Identity - a user's unique ID in GSM/GPRS networks.

**Interface**

Well standardized point in the GPRS standard that typically has multivendor capability; opposite of reference point.

**Internal Network**

Computers and resources protected by the Firewall and accessed by authenticated users.

**IPS**

Check Point Software Blade on a Security Gateway that inspects and analyzes packets and data for numerous types of risks (Intrusion Prevention System).

**IPsec VPN**

Check Point Software Blade on a Security Gateway that provides a Site to Site VPN and Remote Access VPN access.

**ISP**

Internet Service Provider - an organization or operator that sells Internet access.

**J**

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**Jumbo Hotfix Accumulator**

Collection of hotfixes combined into a single package. Acronyms: JHA, JHF, JHFA.

**K**

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**Kerberos**

An authentication server for Microsoft Windows Active Directory Federation Services (ADFS).

**L**

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**LLC**

Logical Link Control - the protocol layer between MS and SGSN.

**Log Server**

Dedicated Check Point server that runs Check Point software to store and process logs.

**Logging & Status**

Check Point Software Blade on a Management Server to view Security Logs from the managed Security Gateways.

**LTE**

Long Term Evolution - a standard for wireless broadband communication for mobile devices and data terminals, based on the GSM/EDGE and UMTS/HSPA technologies. It increases the capacity and speed using a different radio interface together with core network improvements.

**M**

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**Management Interface**

(1) Interface on a Gaia Security Gateway or Cluster member, through which Management Server connects to the Security Gateway or Cluster member. (2) Interface on Gaia computer, through which users connect to Gaia Portal or CLI.

**Management Server**

Check Point Single-Domain Security Management Server or a Multi-Domain Security Management Server.

**Manual NAT Rules**

Manual configuration of NAT rules by the administrator of the Check Point Management Server.

**MIB**

Management Information Base - a collection of managed objects defined by their attributes and visible to the network management system.

**MME**

Mobility management element - in charge of mobility management in GTPv2

**MMS**

Multimedia Short Message Service - wireless service that transmits text, audio and video over WAP.

**Mobile Access**

Check Point Software Blade on a Security Gateway that provides a Remote Access VPN access for managed and unmanaged clients. Acronym: MAB.

**MS**

Mobile Station - a portable device that connects subscribers to a wireless network, for example a cellular phone or a laptop with a cellular modem.

**MS-ISDN**

Mobile Station International ISDN Number - the standard international telephone number used to identify a given subscriber.

**MTP2**

Message Transfer Part layer 2 - S7 protocol layer 2.

**MTP3**

Message Transfer Part layer 3 - SS7 protocol layer 3.

**Multi-Domain Log Server**

Dedicated Check Point server that runs Check Point software to store and process logs in a Multi-Domain Security Management environment. The Multi-Domain Log Server consists of Domain Log Servers that store and process logs from Security Gateways that are managed by the corresponding Domain Management Servers. Acronym: MDLS.

**Multi-Domain Server**

Dedicated Check Point server that runs Check Point software to host virtual Security Management Servers called Domain Management Servers. Synonym: Multi-Domain Security Management Server. Acronym: MDS.

**N**

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**N-Byte**

Number of Bytes.

**N-PDU**

Number of Packets.

**Network Object**

Logical object that represents different parts of corporate topology - computers, IP addresses, traffic protocols, and so on. Administrators use these objects in Security Policies.

**Network Policy Management**

Check Point Software Blade on a Management Server to manage an on-premises environment with an Access Control and Threat Prevention policies.

**NS**

Network Service - the protocol layer between BSS and SGSN.

**NSAPI**

Network Service Access Point Identifier - an integer value in the range [0; 15], used in GTP V0/V1 versions for PDP Context identification in the MS and SGSN.

**NSS**

Network SubSystem - the network part of the network (in GPRS this means SGSN and GGSN).

**O**

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**Open Server**

Physical computer manufactured and distributed by a company, other than Check Point.

**P**

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**P-TMSI**

Packet TMSI - a packet system's temporary mobile's identity.

**PCU**

Packet Control Unit - functional element in BSS that handles upper level GPRS control in radio.

**PDA**

Personal Digital Assistant- a device that fits in hand and has limited services.

**PDN**

Packet Data Network - a network that carries user data in packets (for example, Internet and X.25)

**PDP**

Packet Data Protocol - a network protocol used by an external packet data network (usually IP).

**PDP address**

The MS's address in the external packet data network, also called End User IP address.

**PDP context**

Information sets held in MS and GSNs for a specific PDP address.

**PDU**

Protocol Data Unit - a packet.

**PGW**

Packet Data Network Gateway - an LTE support node.

**PLMN**

Public Land Mobile Network.

**PPP**

Point-to-Point Protocol - a widely used protocol under IP to connect (for example, PC and ISP via modems).

**Provisioning**

Check Point Software Blade on a Management Server that manages large-scale deployments of Check Point Security Gateways using configuration profiles. Synonyms: SmartProvisioning, SmartLSM, Large-Scale Management, LSM.

**PSWT**

Public Switched Telephone Network. A collection of public circuit-switched telephone network, including telephone lines, fixed lines, microwave transmission links, cellular networks, and satellite communication.

**PTM**

Point To Multipoint - one sender, multiple receivers.

**PTP**

Point To Point- one sender, one receiver.

## Q

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### QoS

Check Point Software Blade on a Security Gateway that provides policy-based traffic bandwidth management to prioritize business-critical traffic and guarantee bandwidth and control latency.

## R

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### R

Reference point between a non-ISDN compatible TE and MT. Typically this reference point supports a standard serial interface.

### RA

Routing Area - a set of cells belonging to one group. RA is always a subset of a LA (location area).

### RLC

Radio Link Control - A protocol between MS and BSS to handle retransmission and other radio related issues.

### Rule

Set of traffic parameters and other conditions in a Rule Base (Security Policy) that cause specified actions to be taken for a communication session.

### Rule Base

All rules configured in a given Security Policy. Synonym: Rulebase.

## S

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### S1-MME

Interface between eNodeB and MME.

### S1-U

Interface between eNodeB and SGW.

### S3

Interface between SGSN and MME.

**S4**

Interface between SGSN and SGW.

**S5/S8**

The interface between SGW to PGW on the HPLMN and between PLMNs

**SCTP**

Stream Control Transmission Protocol, SCTP was defined as a transport protocol for SS7 messages to be transmitted over IP networks.

**SecureXL**

Check Point product on a Security Gateway that accelerates IPv4 and IPv6 traffic that passes through a Security Gateway.

**Security Gateway**

Dedicated Check Point server that runs Check Point software to inspect traffic and enforce Security Policies for connected network resources.

**Security Management Server**

Dedicated Check Point server that runs Check Point software to manage the objects and policies in a Check Point environment within a single management Domain. Synonym: Single-Domain Security Management Server.

**Security Policy**

Collection of rules that control network traffic and enforce organization guidelines for data protection and access to resources with packet inspection.

**SGSN**

Serving GSN - a GPRS Support Node.

**SGW**

Serving Gateway - a LTE support node.

**SIC**

Secure Internal Communication. The Check Point proprietary mechanism with which Check Point computers that run Check Point software authenticate each other over SSL, for secure communication. This authentication is based on the certificates issued by the ICA on a Check Point Management Server.

**SLIP**

Serial Line IP protocol - a protocol similar to PPP.

**SM-SC**

Short Message Service Center - a computer that handles short messages.

**SmartConsole**

Check Point GUI application used to manage a Check Point environment - configure Security Policies, configure devices, monitor products and events, install updates, and so on.

**SmartDashboard**

Legacy Check Point GUI client used to create and manage the security settings in versions R77.30 and lower. In versions R80.X and higher is still used to configure specific legacy settings.

**SmartProvisioning**

Check Point Software Blade on a Management Server (the actual name is "Provisioning") that manages large-scale deployments of Check Point Security Gateways using configuration profiles. Synonyms: Large-Scale Management, SmartLSM, LSM.

**SmartUpdate**

Legacy Check Point GUI client used to manage licenses and contracts in a Check Point environment.

**SMS**

Short Message Service - A protocol enabling mobile phone users to send and receive short messages of up to 160 characters messages.

**SMS-GMSC**

Short Message Service Gateway MSC - an MSC used to deliver data to/from SGSN.

**SMS-IWMSC**

Short Message Service Interworking MSC - an MSC used to deliver data to/from SGSN.

**SNDC**

SubNetwork Dependent Convergence - The protocol layer between MS and SGSN.

**SNDCP**

SubNetwork Dependent Convergence Protocol - the protocol used in SNDC.

**SNMP**

Simple Network Management Protocol runs over TCP/IP and is used to control and manage IP gateways and other network functions.

**Software Blade**

Specific security solution (module): (1) On a Security Gateway, each Software Blade inspects specific characteristics of the traffic (2) On a Management Server, each Software Blade enables different management capabilities.

**Standalone**

Configuration in which the Security Gateway and the Security Management Server products are installed and configured on the same server.

**T**

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**T-PDU**

An original packet from an MS or a network node in an external packet data network.

**TCAP**

Transaction Capabilities Application Part - SS7 protocol layer.

**TE**

Terminal Equipment - typically a computer, host.

**TEID**

Tunnel End Point Identification - The GTP version 1 uni-directional tunnel identifier.

**TFT**

Traffic Flow Template, a packet filter list that sorts the packets coming into the GGSN to the correct PDP Context. Also allows some protocol security filtering.

**Threat Emulation**

Check Point Software Blade on a Security Gateway that monitors the behavior of files in a sandbox to determine whether or not they are malicious. Acronym: TE.

**Threat Extraction**

Check Point Software Blade on a Security Gateway that removes malicious content from files. Acronym: TEX.

**TID**

Tunnel ID - the GTP version 0 GTP tunnel identifier. Consists of the user ID, or equivalent when Anonymous Access is used, and NSAPI.

**TLLI**

Temporary Logical Link Identity - provides a signaling address for communication between the MS and the SGSN.

**U**

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**Um**

Radio interface between MS and the network.

**UMTS**

Universal Mobile Telephone System, a third generation service (part of the IMT-2000 vision) that is expected to enable cellular service providers to deliver high-value broadband information, commerce and entertainment services to mobile users via fixed, wireless and satellite networks.

**Updatable Object**

Network object that represents an external service, such as Microsoft 365, AWS, Geo locations, and more.

**URL Filtering**

Check Point Software Blade on a Security Gateway that allows granular control over which web sites can be accessed by a given group of users, computers or networks. Acronym: URLF.

**User Directory**

Check Point Software Blade on a Management Server that integrates LDAP and other external user management servers with Check Point products and security solutions.

**UTMS**

Universal Mobile Telecommunications System. A third generation, packet-based, mobile cellular technology for networks based on the GSM standard.

## V

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### **VPLMN**

Visited Public Land Mobile Network - the network where the MS is currently located.

### **VSX**

Virtual System Extension. Check Point virtual networking solution, hosted on a computer or cluster with virtual abstractions of Check Point Security Gateways and other network devices. These Virtual Devices provide the same functionality as their physical counterparts.

### **VSX Gateway**

Physical server that hosts VSX virtual networks, including all Virtual Devices that provide the functionality of physical network devices. It holds at least one Virtual System, which is called VS0.

## W

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### **WAP**

Wireless Application Protocol, a standard wireless protocol specification, based on existing Internet standards such as XML and IP, that leverages HTTP and enables developers to use existing tools to produce scalable applications that deliver Internet content and advanced services to mobile phones and other wireless terminals.

## Z

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### **Zero Phishing**

Check Point Software Blade on a Security Gateway (R81.20 and higher) that provides real-time phishing prevention based on URLs. Acronym: ZPH.